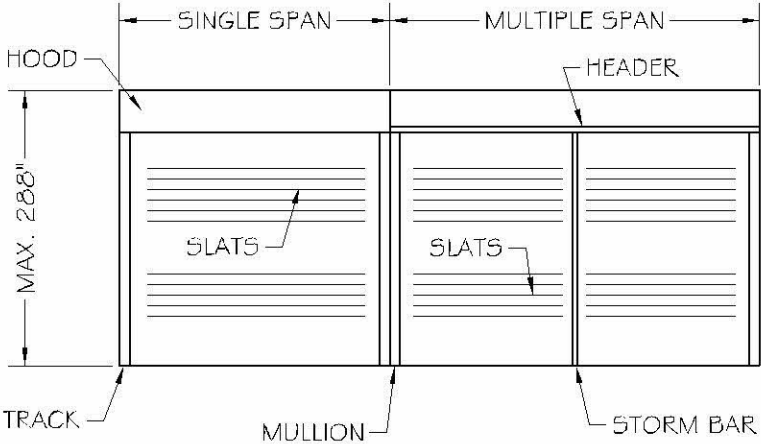
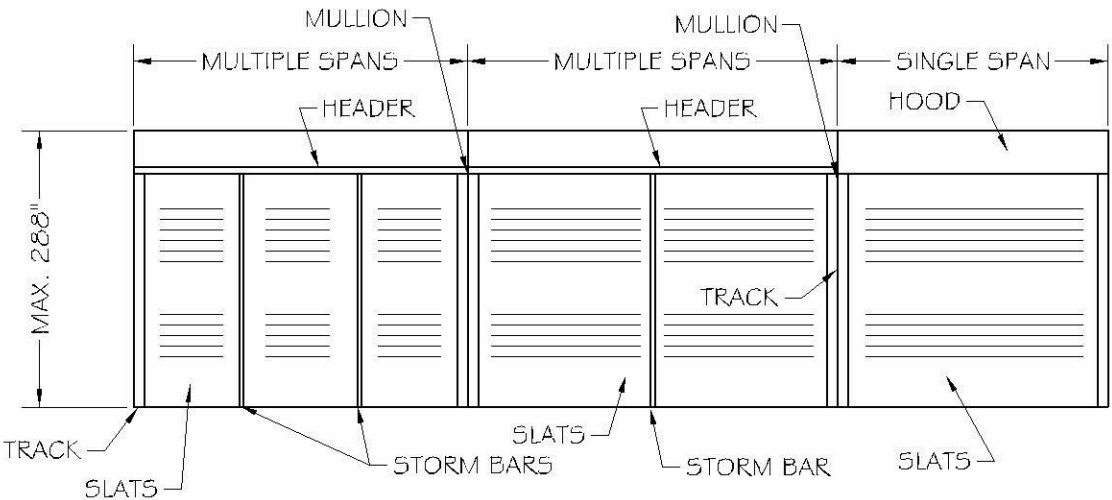
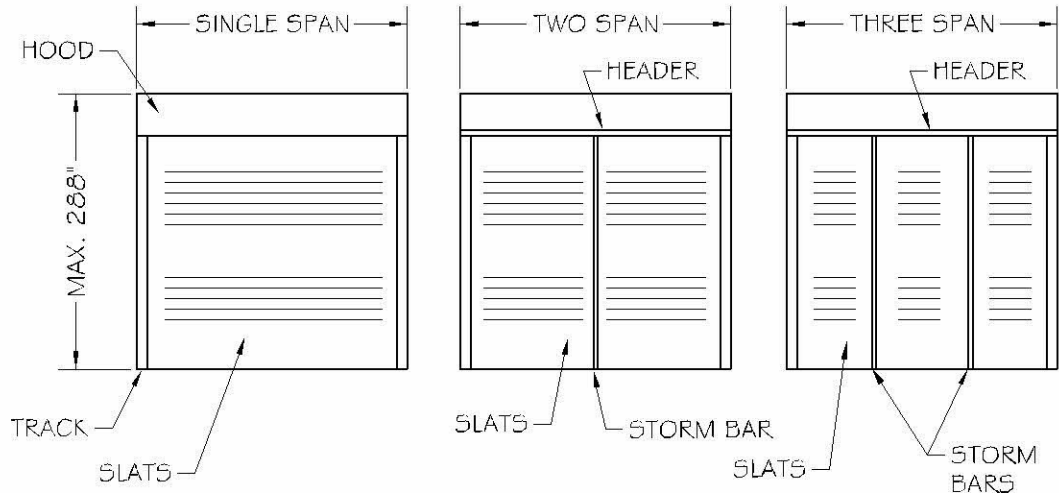


GENERAL NOTES:
(FLORIDA BUILDING CODE 2023. 8th EDITION - non HVHZ)

- 1- DEFINITION: THIS PRODUCT IS A HURRICANE PROTECTION DEVICE; DESIGNED, CONSTRUCTED AND ERECTED TO EASILY ENCLOSE AN AREA, PROVIDING PROTECTION FROM HURRICANE FORCE WINDS AND WIND BORNE DEBRIS WITHIN THE ALLOWABLE DESIGNED PRESSURES AND LIMITATIONS STATED IN THIS APPROVAL.
- 2- POSTING: A PERMANENT LABEL SHALL BE PROVIDED AS PER FBC AND IT SHALL BE LOCATED ON THE BOTTOM OF THE HOOD FACING THE EXTERIOR OR OUTSIDE AND A MINIMUM OF ONE LABEL PER OPENING.
- 3- GLASS SEPARATION (GS): PRODUCT IS QUALIFIED AS "NON POROUS", THEREFORE GLAZING SEPARATION IS NOT REQUIRED WITH EXCEPTION FOR THE FOLLOWING LOCATIONS:
3.1- WIND ZONE 4 AS DEFINED IN ASTM E 1996 (LOCATIONS WITH ULTIMATE WIND SPEED GREATER THAN 170 MPH AS PER FBC SECTION 1609.1.2.2)
3.2- ESSENTIAL FACILITIES AS DEFINED IN ASCE 7
FOR THE ABOVE LOCATIONS THE PRODUCT MUST BE SEPARATED FROM GLAZING AREAS AS PER TABLE 1.
- 4- LOADS: DESIGNED LOAD CALCULATED BASED ON THE ASCE 7-22 AND PROVIDED BY A PROFESSIONAL ARCHITECT OR ENGINEER FOR EACH SPECIFIC PROJECT. THE CALCULATED DESIGNED PRESSURE MUST NOT EXCEED THE ALLOWABLE PRESSURES FOR EACH SHUTTER COMPONENT TO BE USED. THE DESIGN PRESSURES ARE PERMITTED TO BE MULTIPLIED BY 0.6
- 5- MATERIAL: ALL EXTRUDED ALUMINUM SHAPES SHALL BE MADE OF 6063-T6 OR AS NOTED.
- 6- FASTENERS: ASSEMBLY SCREWS AND ANCHORS SHALL BE AS SPECIFIED IN THE CURRENT SET OF DRAWINGS. INSTALLATION AND LOADS AS PER THIS APPROVAL.
- 7- MULTIPLE SPAN SHUTTERS: REINFORCEMENTS SUCH AS STORM BARS, HEADER AND MULLIONS WERE DESIGNED BY RATIONAL ANALYSIS AND ARE PART OF THESE DRAWINGS. DUE TO MANY VARIATIONS OF REINFORCEMENTS, LOADS AND SHUTTER DIMENSIONS BECOME IMPRACTICAL ADDRESSING ALL CONDITIONS IN ONE SET OF DRAWINGS. THE CONDITIONS ADDRESSED IN THE PROPOSED DRAWINGS HAVE CONSERVATIVE LIMITATIONS IN ORDER TO FACILITATE THE DESIGN. SITE SPECIFIC CONDITIONS CAN BE ADDRESSED ON A JOB BASIS PROVIDED PROPER ENGINEERING.
- 8- USE: IT SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR, ARCHITECT OR ENGINEER OF RECORD TO VERIFY THE FOLLOWING:
8.1- THE STABILITY OF THE STRUCTURE WHERE THE SHUTTER IS TO BE ATTACHED INSURING PROPER ANCHORAGE.
8.2- THE SITE SPECIFIC PROJECT CRITERIA, SUCH AS BUT NOT LIMITED TO, WIND LOADS, LOCAL CODE REQUIREMENTS, DESIGNED PRESSURES ETC.
8.3- THAT THIS APPROVAL IS ADEQUATE TO THE SPECIFIC PROJECT.
- 9- LIMITATIONS: NOT DESIGNED FOR HIGH VELOCITY HURRICANE ZONES HVHZ (MIAMI-DADE AND BROWARD COUNTIES)

CODE AND STANDARD COMPLIANCE:
FLORIDA BUILDING CODE 2023 - 8th EDITION (non HVHZ)
ASTM E330, ASTM E1886, ASTM E1996 (LEVEL D MISSILE)
ASCE 7-22, ADM-20, NDS 2018

TYPICAL SHUTTER CONFIGURATIONS



DESIGNED PRESSURE	
COMPONENT	REF. SHEET
SLAT DESIGN	2
TRACK ANCHORAGE	3-4
STORM BAR	5-8
HEADER	9-10
MULLION	11-17

ALLOWABLE DESIGNED PRESSURE FOR SHUTTERS ARE FUNCTION OF THE LOWEST ALLOWABLE DESIGNED LOAD FOR THE SLATS, TRACK ANCHORAGE, STORM BARS, HEADERS, MULLIONS.

USE REFERENCED SHEET ABOVE TO CALCULATED THE THE DESIGNED LOAD FOR EACH OF THE COMPONENTS.



Pedro De Figueiredo
PE 52609
09/16/2023

Pedro
Figueiredo

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Pedro Figueiredo
Date: 2023.09.16
11:04:24 -04'00'

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THE DATE INDICATED HERE.

ENGINEERING SERVICES
CA 82439
9715 W. BROWARD BLVD. 247
PLANTATION, FL 33324
PFMIF@AOL.COM



PRODUCT:
E555 - E544
EXTRUDED ALUMINUM ROLLING SHUTTER
LARGE & SMALL MISSILE
IMPACT RATED

MANUFACTURER:
CROCI
NORTH AMERICA
11600 ADELMO LANE FORT MYERS, FL 33966
TEL.: (239) 278-3066 - FAX: (239) 278-5977
EMAIL: INFO@CROCIUSA.COM

SCALE:

NA

DATE:

09/16/2023

DRAWING #

23-165

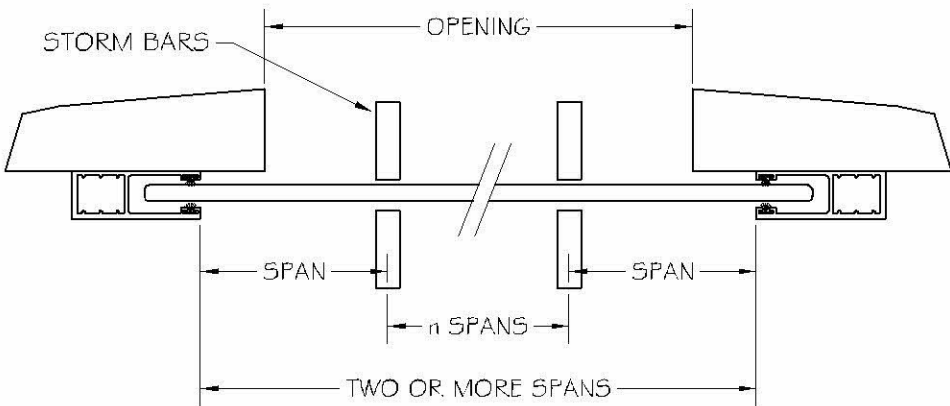
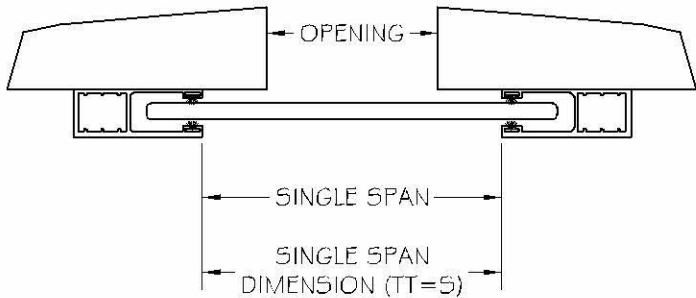
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1 of 18

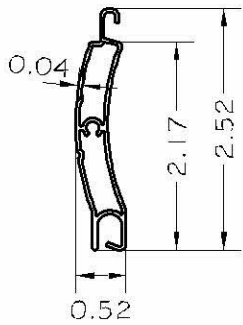
TABLE 1 - SLAT DESIGN

ALLOWABLE PRESSURE ± PSF	ES 55 SLAT		ES 44 SLAT	
	MAX. SPAN	MIN. G5	MAX. SPAN	MIN. G5
40	80	5.75	51	5.50
50	74	5.75	48	5.50
60	70	5.25	44	5.00
70	66	4.75	41	5.00
80	62	4.50	38	5.00
90	58	4.50	36	5.00
100	54	4.50	34	5.00
110	49	4.50	32	5.00
120	45	4.50	30	5.00
130	42	4.50	28	5.00
140	39	4.50	26	5.00
150	36	4.50	24	5.00

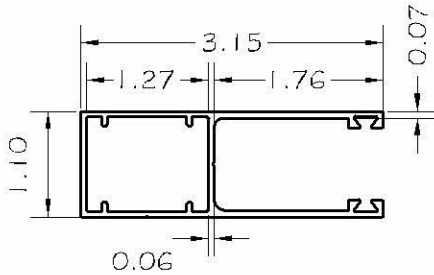
1- ALLOWABLE PRESSURE IN PSF
2- MAX. SPAN: MAXIMUM SLAT SPAN IN INCHES
3- MIN. G5: MINIMUM GLAZING SEPARATION ONLY APPLICABLE AS PER GENERAL NOTE 3



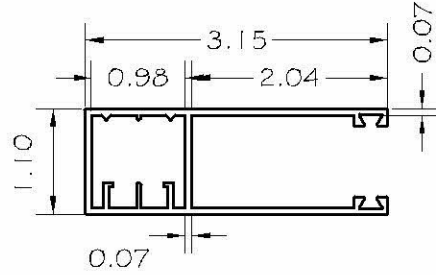
PRODUCT COMPONENTS



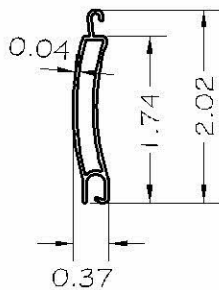
A1- E555
SLAT BLADE
6063-T6 Alloy



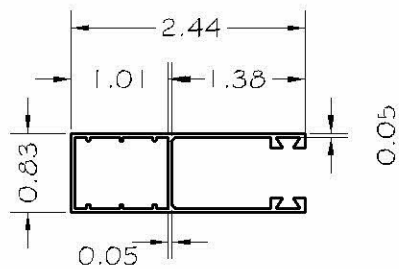
A2- GE458028 SIDE RAIL
COMPATIBLE WITH E555,
6063-T6 ALLOY



A21- GE8028 ALTERNATE SIDE RAIL
COMPATIBLE WITH E555 6063-T6
ALLOY

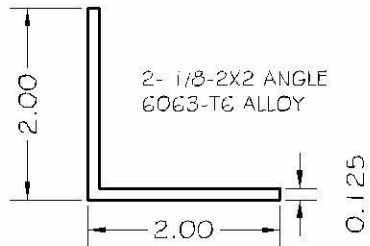


B1- E544
SLAT BLADE
6063-T6 Alloy

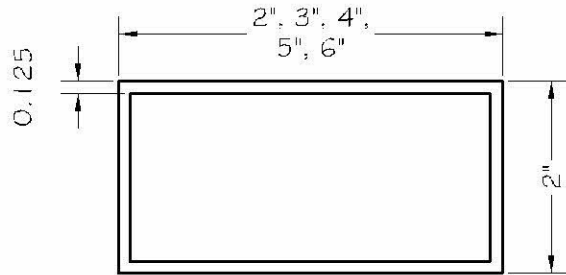


B2- GE62X21 SIDE RAIL
COMPATIBLE WITH E544
6063-T6 ALLOY

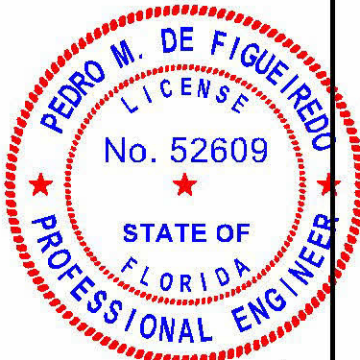
1- TRACK FELT
FELT PILE



2- 1/8-2X2 ANGLE
6063-T6 ALLOY



3- 1/8-2x ALUMINUM TUBE
6063-T6 Alloy



Pedro De Figueiredo
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PRODUCT:
E555 - ES44
EXTRUDED ALUMINUM ROLLING SHUTTER
LARGE & SMALL MISSILE
IMPACT RATED

MANUFACTURER:
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11600 ADELMO LANE FORT MYERS, FL 33966
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E 55 - GUIDE RAILS MOUNTING OPTIONS



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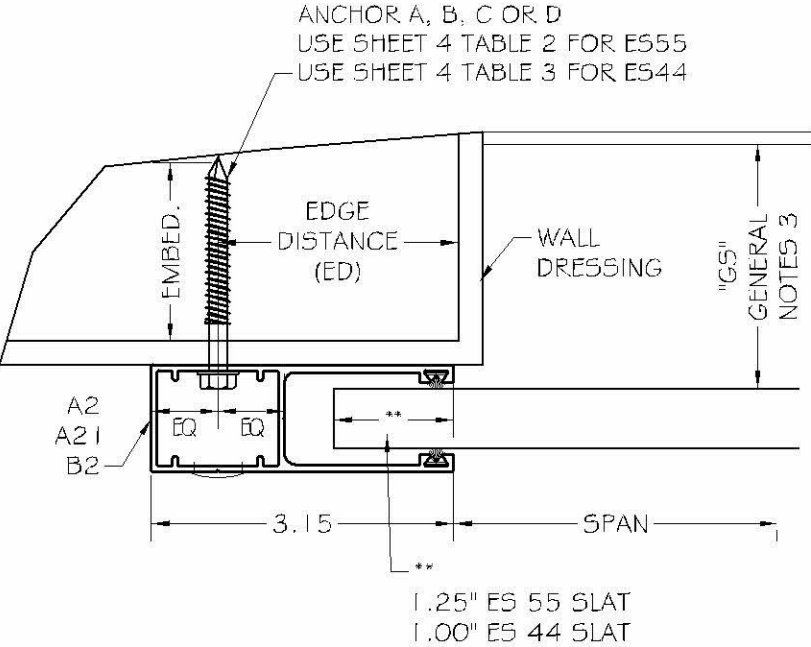


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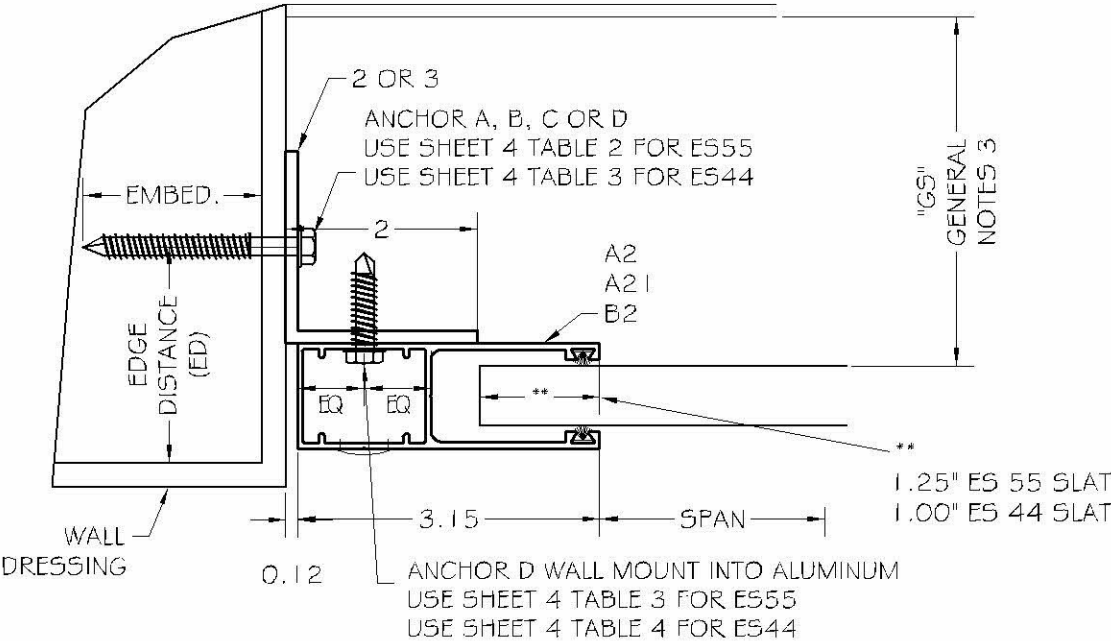
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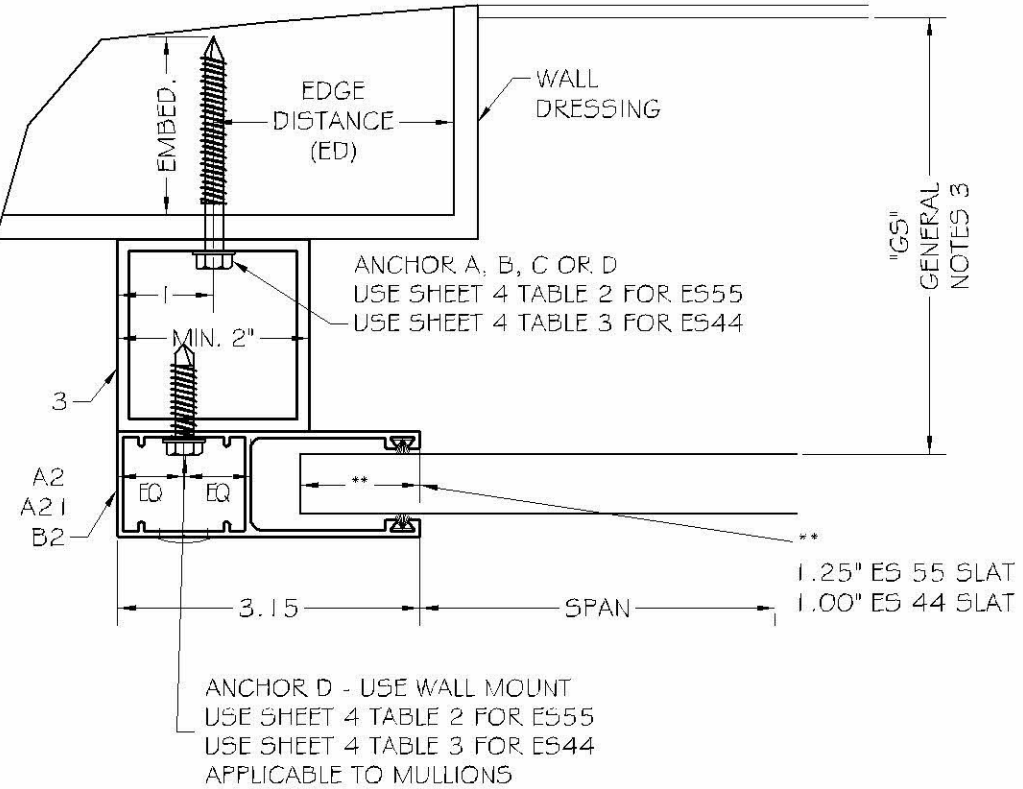
A- WALL MOUNT



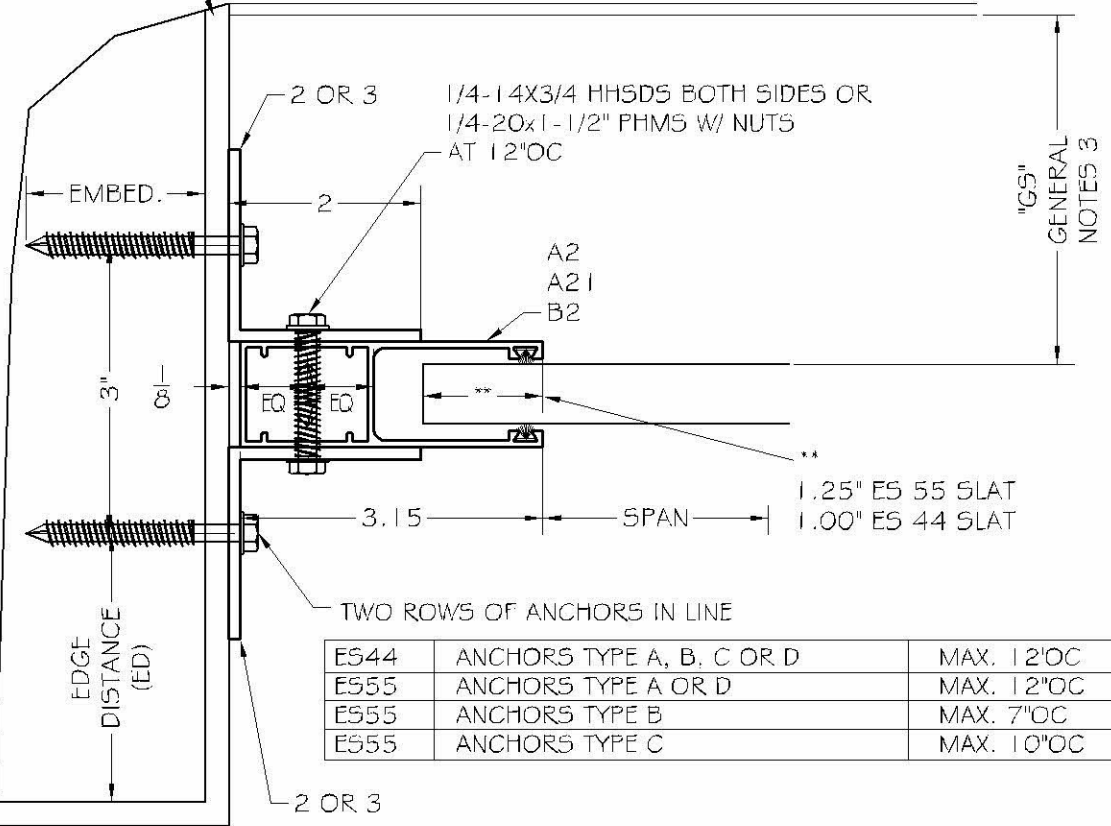
C- INSET MOUNT



B- BUILT-OUT MOUNT



D- ALTERNATE INSET MOUNT



E544	ANCHORS TYPE A, B, C OR D	MAX. 12"OC
E555	ANCHORS TYPE A OR D	MAX. 12"OC
E555	ANCHORS TYPE B	MAX. 7"OC
E555	ANCHORS TYPE C	MAX. 10"OC

GUIDE RAILS ANCHORAGE DESIGN

TABLE 2 - ES 55 MAXIMUM TRACK ANCHORAGE SPACING

Pd	SPAN	GE458028 RAIL WALL MOUNT ANCHOR TYPE				BUILT-OUT MOUNT ANCHOR TYPE				INSET MOUNT ANCHOR TYPE				GE8028 RAIL WALL MOUNT ANCHOR TYPE			
		A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
40	80	5	3	5	5	12	5	12	11	9	4	6	8	4	3	4	4
	60	7	5	7	7	12	7	12	12	12	5	9	11	6	4	6	6
	40	11	7	11	11	12	11	12	12	12	8	12	12	9	6	9	9
	30	12	10	12	12	12	12	12	12	12	11	12	12	12	8	12	12
60	80	3	-	3	3	8	3	8	7	6	-	4	5	3	-	3	3
	60	5	3	5	5	11	5	11	9	8	3	6	7	4	-	4	4
	50	6	4	6	6	12	6	12	11	10	4	7	9	5	3	5	5
	40	7	5	7	7	12	7	12	12	12	5	9	11	6	4	6	6
80	30	10	7	10	10	12	10	12	12	12	7	12	12	8	5	8	8
	80	-	-	-	-	6	-	6	5	4	-	3	4	-	-	-	-
	60	3	-	3	3	8	3	8	7	6	-	4	5	3	-	3	3
	50	4	3	4	4	10	4	10	8	7	3	5	7	3	-	3	3
100	40	5	3	5	5	12	5	12	11	9	4	6	8	4	3	4	4
	30	7	5	7	7	12	7	12	12	12	5	9	11	6	4	6	6
	60	3	-	3	3	6	3	6	5	5	-	3	4	-	-	-	-
	50	3	-	3	3	8	3	8	7	6	-	4	5	3	-	3	3
110	30	6	4	6	6	12	6	12	11	10	4	7	9	5	3	5	5
	50	3	-	3	3	7	3	7	6	5	-	4	5	-	-	-	-
	40	4	-	4	4	9	4	9	8	6	3	5	6	3	-	3	3
	30	5	3	5	5	12	5	12	10	9	4	6	8	4	3	4	4
130	50	-	-	-	-	6	-	6	5	4	-	3	4	-	-	-	-
	40	3	-	3	3	7	3	7	6	5	-	4	5	-	-	-	-
	30	4	3	4	4	10	4	10	9	7	3	5	7	3	-	3	3
150	40	3	-	3	3	5	3	5	5	5	-	3	4	-	-	-	-
	36	3	-	3	3	6	3	6	6	5	-	4	5	-	-	-	-
	30	4	-	4	4	7	4	7	7	6	3	4	6	3	-	3	3

TABLE 3 - ES 44 MAXIMUM TRACK ANCHORAGE SPACING

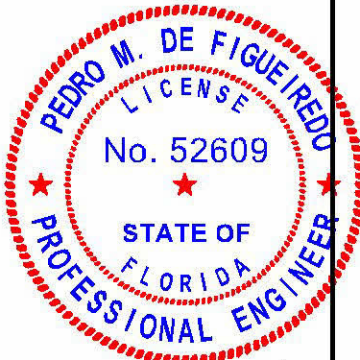
Pd	SPAN	WALL MOUNT ANCHOR TYPE				BUILT-OUT MOUNT ANCHOR TYPE				INSET MOUNT ANCHOR TYPE			
		A	B	C	D	A	B	C	D	A	B	C	D
40	51	6	6	6	6	12	11	12	12	12	8	12	12
	42	8	7	8	8	12	12	12	12	12	10	12	12
	32	10	9	10	10	12	12	12	12	12	12	12	12
	24	12	12	12	12	12	12	12	12	12	12	12	12
60	48	4	4	4	4	9	8	9	9	9	6	8	9
	44	5	4	5	5	9	8	9	9	9	6	9	9
	42	5	4	5	5	10	9	10	10	10	6	10	10
	32	7	6	7	7	12	12	12	12	12	9	12	12
80	24	9	8	9	9	12	12	12	12	12	12	12	12
	48	3	3	3	3	6	6	6	6	6	4	6	6
	44	3	3	3	3	7	6	7	7	7	4	7	7
	42	4	3	4	4	7	7	7	7	7	5	7	7
100	32	5	4	5	5	10	9	10	10	10	6	9	10
	24	7	6	7	7	12	12	12	12	12	9	12	12
	44	3	-	3	3	6	5	6	6	6	3	5	6
	42	3	-	3	3	6	5	6	6	6	4	6	6
110	32	4	3	4	4	8	7	8	8	8	5	7	8
	24	5	5	5	5	10	9	10	10	10	7	10	10
	42	-	-	-	-	5	5	5	5	5	3	5	5
	32	3	3	3	3	7	6	7	7	7	4	7	7
130	24	5	4	5	5	9	8	9	9	9	6	9	9
	32	3	3	3	3	6	5	6	6	6	4	6	6
	28	3	3	3	3	7	6	7	7	7	4	7	7
	24	4	4	4	4	8	7	8	8	8	5	8	8
150	28	3	-	3	3	6	5	6	6	6	4	6	6
	26	3	3	3	3	6	6	6	6	6	4	6	6
	24	3	3	3	3	7	6	7	7	7	4	7	7

ANCHOR SCHEDULE

TYPE	DESCRIPTION	MANUFACTURER	EMBEDMENT	ED	BASE MATERIAL
A	1/4" 55 TAPCON	ITW	1 3/4"	2 1/2"	3000 psi CONCRETE
B	1/4" 55 TAPCON	ITW	1 1/4"	2 1/2"	HOLLOW BLOCK
C	1/4"-14 55 HH5MS	GENERIC	1 1/2"	1"	WOOD (SG>=.46)
D	1/4"-14 55 HH5MS	GENERIC	1/8"	3/4"	1/8" METAL STRUCTURE

ANCHOR NOTES:

- 1- 1/4" TAPCONS CAN BE USED AS TYPE C INTO WOOD
2- 1/4" CARBON STEEL TAPCONS CAN BE USED AS AN ALTERNATE TO THE STAINLESS STEEL
3- METAL STRUCTURES ARE MADE OF MINIMUM 6063-T6 ALUMINUM OR ASTM A36 STEEL



Pedro De Figueiredo
PE 52609
09/16/2023

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PRODUCT:
ES55 - ES44
EXTRUDED ALUMINUM ROLLING SHUTTER
LARGE & SMALL MISSILE
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SHEET

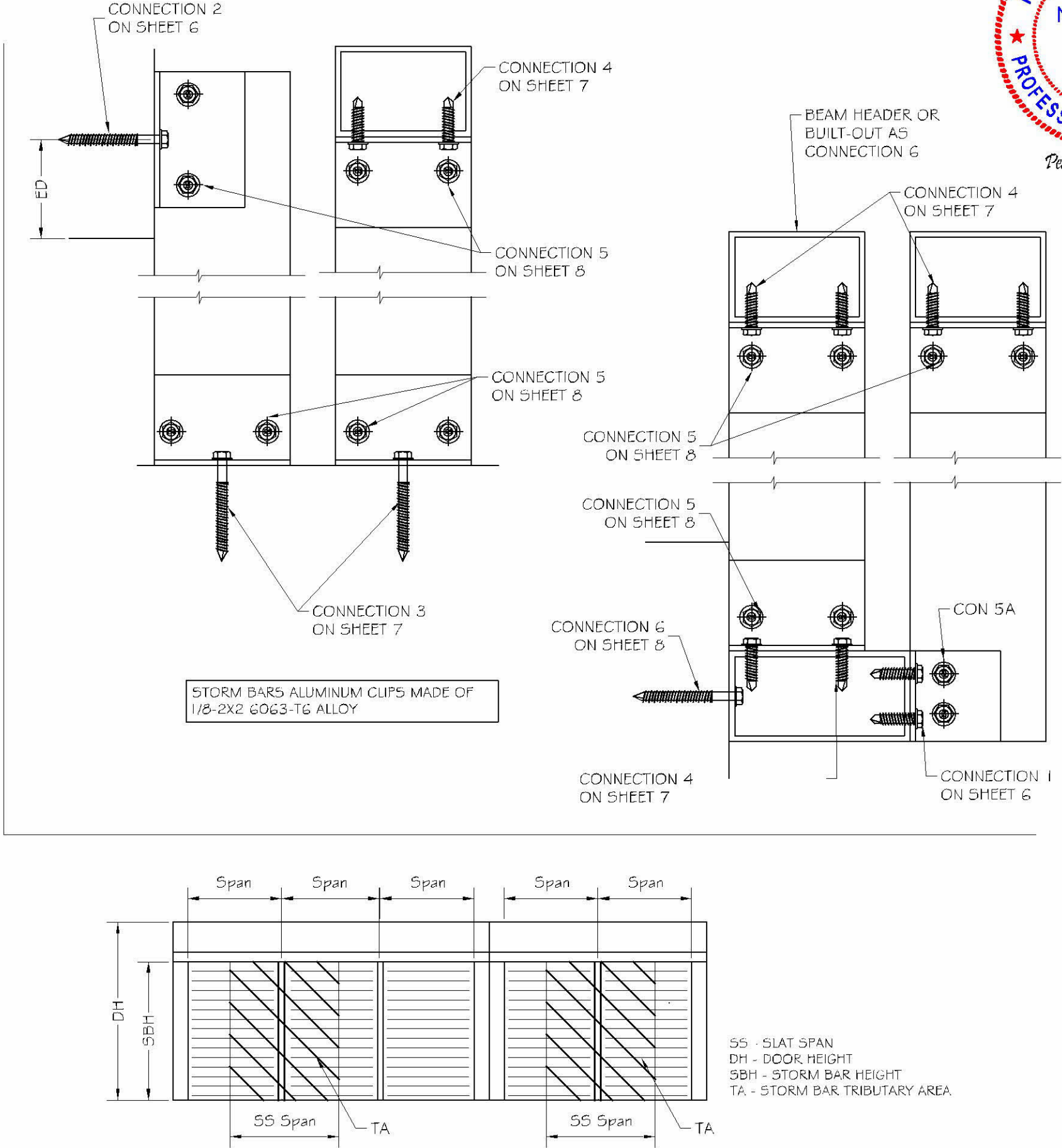
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TABLE 4- STORM BARS
ALLOWABLE DESIGN LOAD - Pd (Psf)

SS	SBH	STORM BAR TYPE				
		2X2	2X3	2X4	2X5	2X6
24	36	150	150	150	150	150
	48	150	150	150	150	150
	60	99	150	150	150	150
	72	57	135	150	150	150
	84	36	96	150	150	150
	96	24	64	114	150	150
	108	--	45	89	124	150
	120	--	32	66	100	131
30	36	150	150	150	150	150
	48	137	150	150	150	150
	60	79	150	150	150	150
	72	45	108	150	150	150
	84	28	76	120	150	150
	96	--	51	91	127	150
	108	--	36	71	99	131
	120	--	26	53	80	105
36	36	150	150	150	150	150
	48	114	150	150	150	150
	60	66	130	150	150	150
	72	38	90	137	150	150
	84	24	64	100	139	150
	96	--	42	76	106	139
	108	--	30	59	83	109
	120	--	21	44	66	87
48	36	110	110	110	110	110
	48	86	110	110	110	110
	60	49	97	110	110	110
	72	28	67	103	110	110
	84	--	48	75	104	110
	96	--	32	57	79	104
	108	--	22	44	62	82
	120	--	--	33	50	65
54	36	100	100	100	100	100
	48	76	100	100	100	100
	60	44	86	100	100	100
	72	25	60	91	100	100
	84	--	42	66	93	100
	96	--	28	50	70	93
	108	--	20	39	55	72
	120	--	--	29	44	58
60	36	85	85	85	85	85
	48	68	85	85	85	85
	60	39	78	85	85	85
	72	22	54	82	85	85
	84	--	38	60	83	85
	96	--	25	45	63	83
	108	--	--	35	49	65
	120	--	--	26	40	52
80	36	40	40	40	40	40
	48	40	40	40	40	40
	60	29	40	40	40	40
	72	--	40	40	40	40
	84	--	28	40	40	40
	96	--	--	34	40	40
	108	--	--	26	37	40
	120	--	--	20	30	39

STORM BAR NOTES:
1- SS: SLAT SPAN. SBH: STORM BAR HEIGHT
2- TABLES MUST WORK IN CONJUNCTION WITH MAXIMUM ALLOWABLE SLAT SPAN TABLES ON SHEET 2.
3- THESE TABLES ARE FURTHER LIMITED BY TYPE OF ANCHORAGE USED.
4- SPECIFIC CONDITIONS OUT OF THE SCOPE OF THESE TABLES CAN BE ADDRESSED WITH A SITE SPECIFIC DESIGN BY A PROFESSIONAL ENGINEER.

STORM BARS MOUNTING OPTIONS



Pedro De Figueiredo
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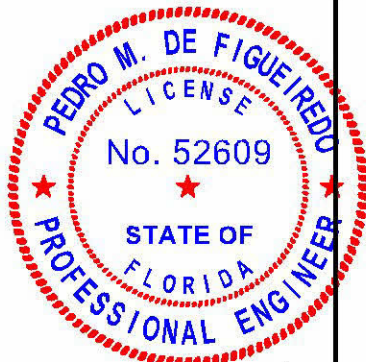
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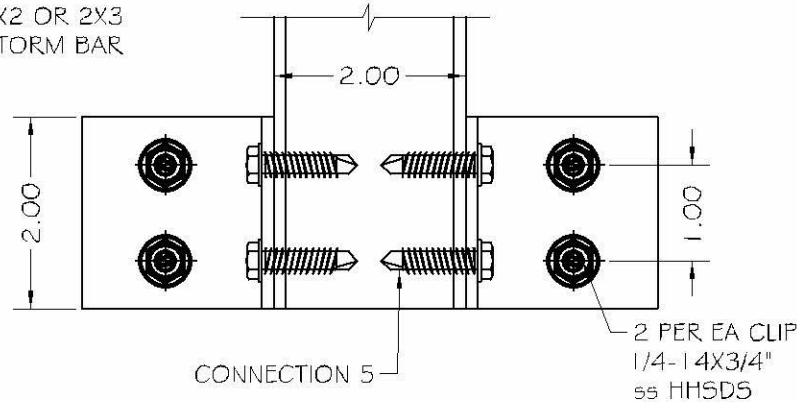
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NORTH AMERICA
11600 ADELMO LANE FORT MYERS, FL 33966
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EMAIL: INFO@CROCIUSA.COM

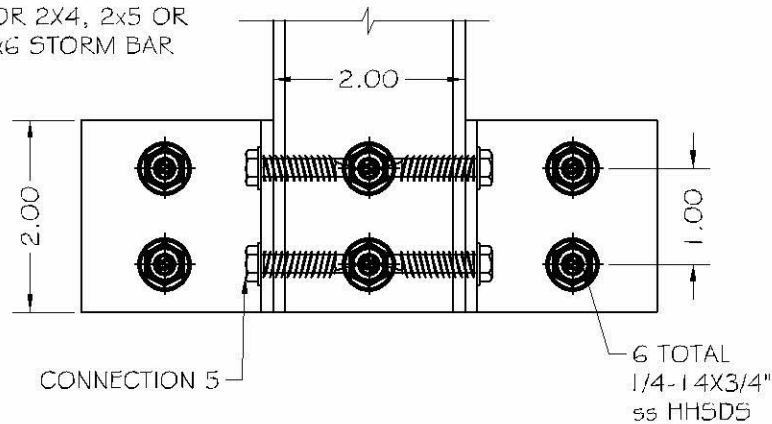
SCALE:
NA
DATE:
09/16/2023
DRAWING #
23-165
SHEET
6 of 18

CONNECTION 1 STORM BARS ANCHORAGE WALL MOUNTED TO ALUMINUM TUBE

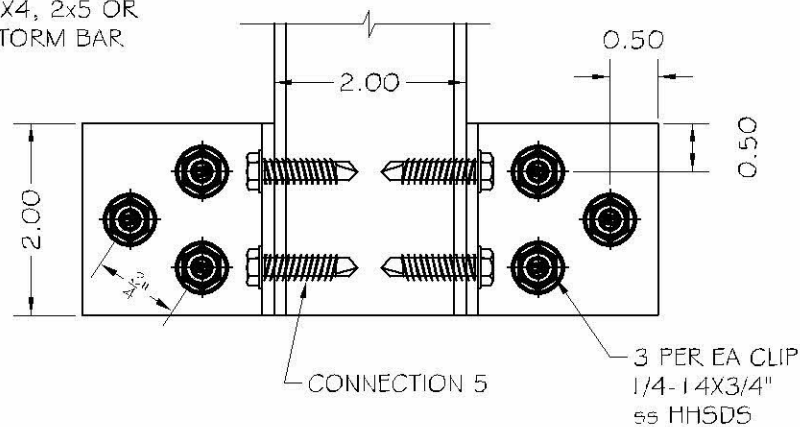
TYPICAL FOR
2X2 OR 2X3
STORM BAR



TYPICAL FOR
FOR 2X4, 2X5 OR
2X6 STORM BAR

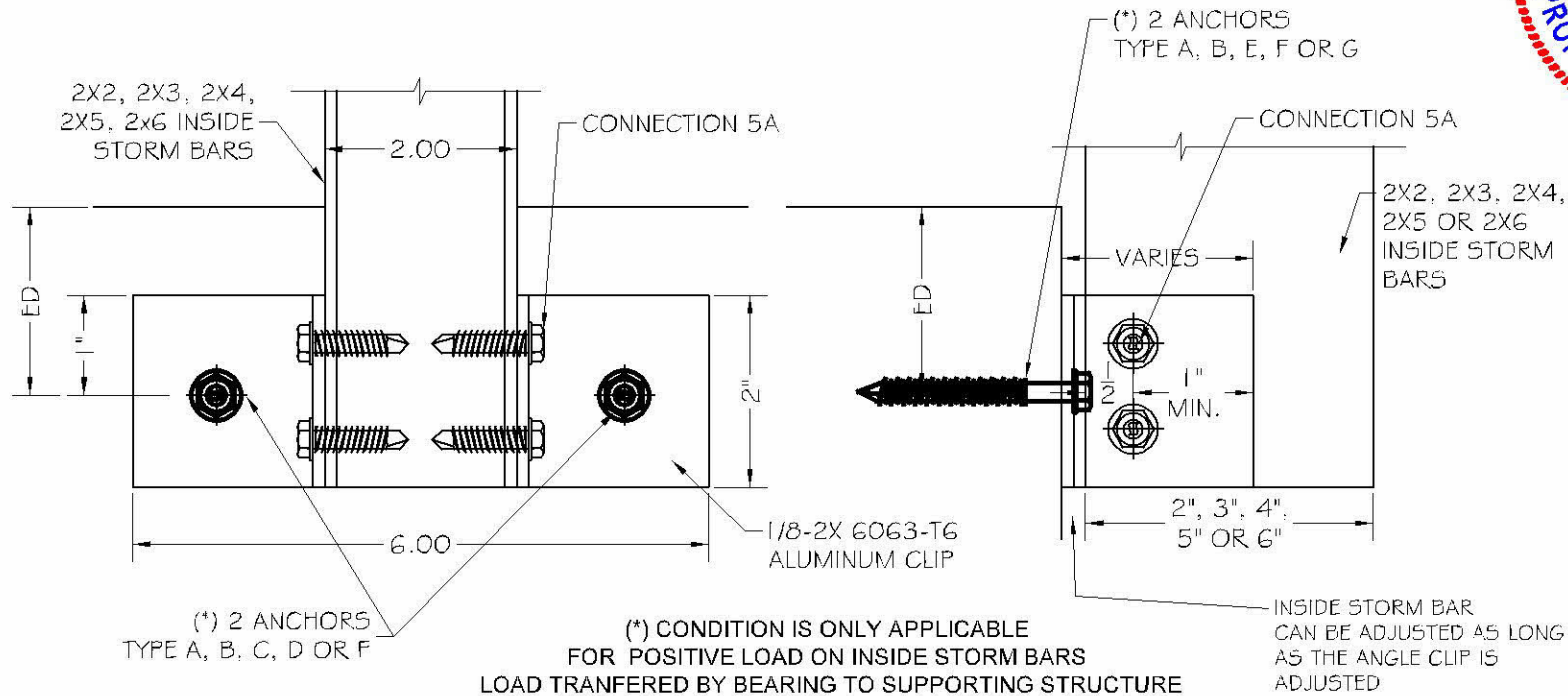


OPTIONAL CONNECTION
FOR 2X4, 2X5 OR
2X6 STORM BAR



CONNECTION CAPACITY IS LIMITED TO THE MAXIMUM ALLOWABLE PRESSURES
SPECIFIED AT TABLE 4 ON SHEET 5

CONNECTION 2 STORM BARS ANCHORAGE TO WALL

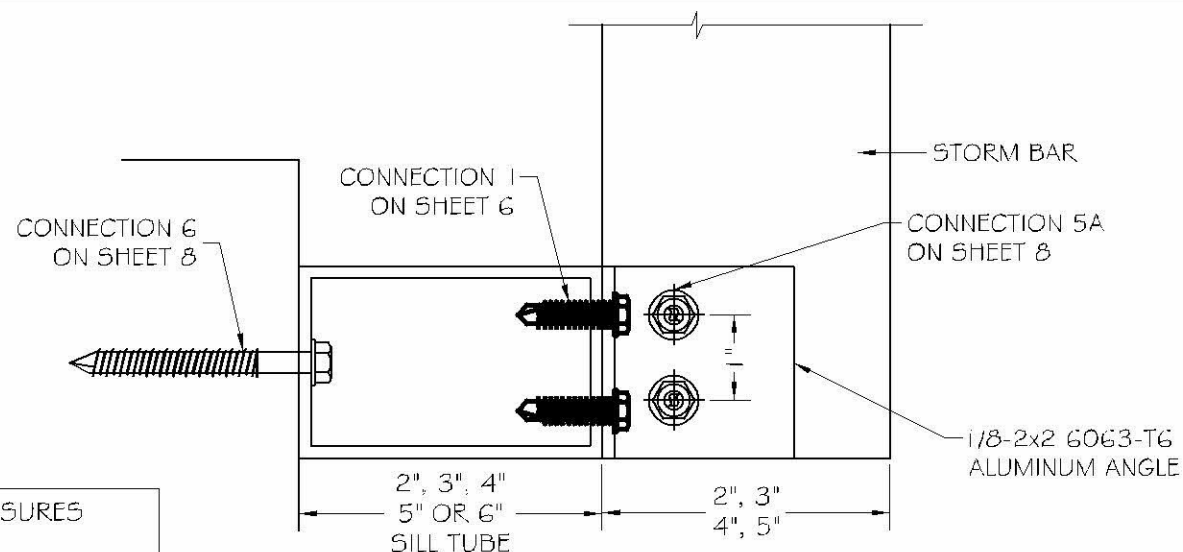


ANCHOR SCHEDULE

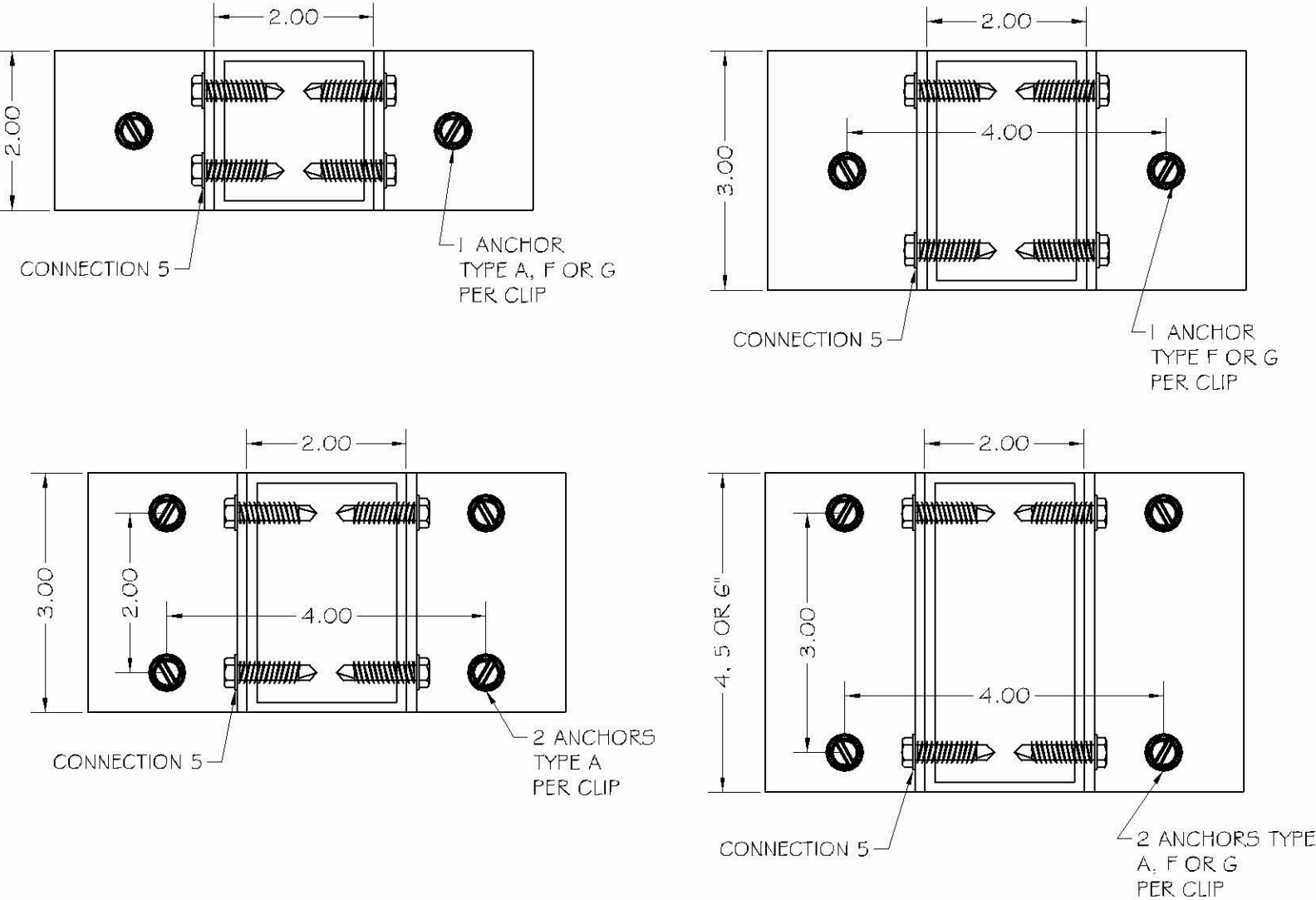
TYPE	DESCRIPTION	MANUFACTURER	EMBEDMENT	ED	BASE MATERIAL
A	1/4" ss TAPCON	ITW	1 3/4"	2 1/2"	3000 psi CONCRETE
B	1/4" ss TAPCON	ITW	1 1/4"	2 1/2"	HOLLOW BLOCK
C	1/4"-14 ss HH5MS	GENERIC	1 1/2"	1"	WOOD (SG > .46)
D	1/4"-14 ss HH5MS	GENERIC	1/8"	3/4"	1/8" METAL STRUCTURE
F	1/4"-20 ss SOLID-SET	ALL POINTS	1"	2 1/2"	3000 psi CONCRETE

ANCHOR NOTES:

- 1- 1/4" TAPCONS CAN BE USED AS TYPE C INTO WOOD
- 2- 1/4" CARBON STEEL TAPCONS CAN BE USED AS AN ALTERNATE TO THE STAINLESS STEEL
- 3- METAL STRUCTURES ARE MADE OF MINIMUM 6063-T6 ALUMINUM OR ASTM A36 STEEL



CONNECTION 3 - STORM BARS ANCHORAGE
FLOOR/CEILING MOUNTED TO CONCRETE



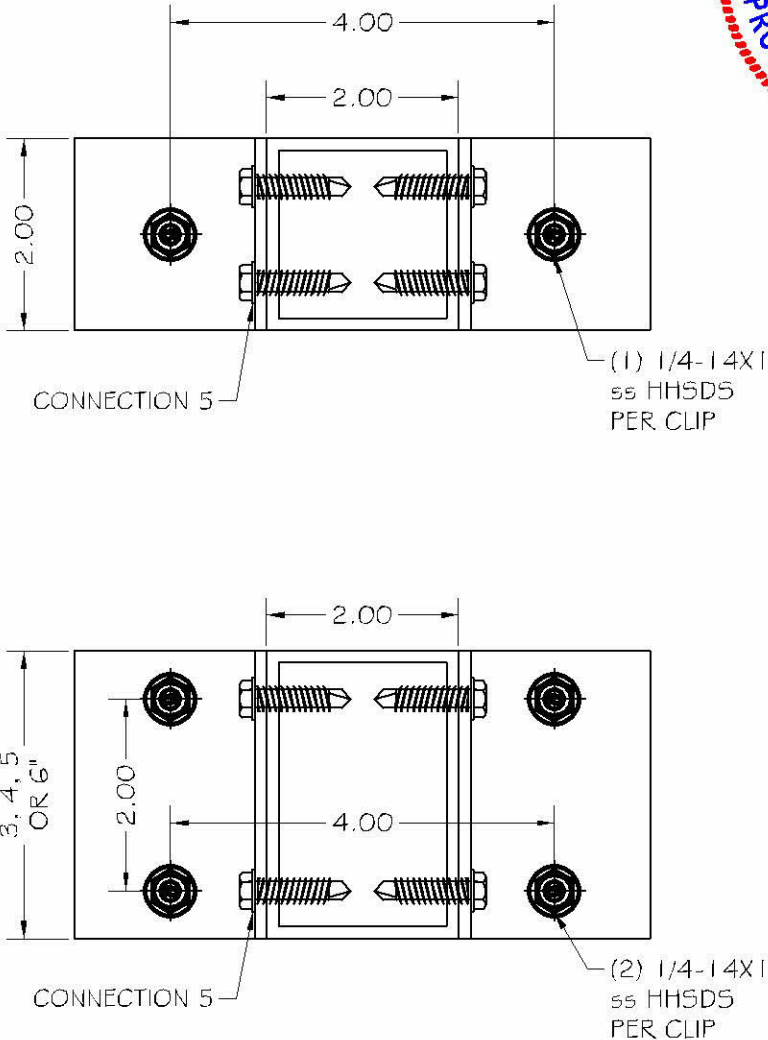
CONNECTION CAPACITY IS LIMITED TO THE MAXIMUM ALLOWABLE PRESSURES
SPECIFIED AT TABLE 4 ON SHEET 5

ANCHOR SCHEDULE

TYPE	DESCRIPTION	MANUFACTURER	EMBEDMENT	ED	BASE MATERIAL
A	1/4" ss TAPCON	ITW	1 3/4"	2 1/2"	3000 psi CONCRETE
F	1/4"-20 ss SOLID-SET	ALL POINTS	1"	2 1/2"	3000 psi CONCRETE
G	5/16" TAPCON	ITW	2 1/4"	2 1/2"	3000 psi CONCRETE

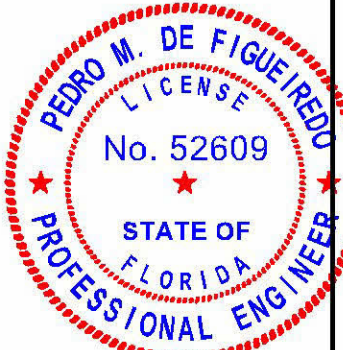
ANCHOR NOTES:
1- 1/4" CARBON STEEL TAPCONS CAN BE USED AS AN ALTERNATE TO THE STAINLESS STEEL

CONNECTION 4
STORM BARS ANCHORAGE FLOOR/CEILING
TO BUILT-OUT TUBE OR HEADER



CONNECTION CAPACITY IS LIMITED TO THE MAXIMUM
ALLOWABLE PRESSURES SPECIFIED AT TABLE 4 ON SHEET 5

ANCHOR NOTES:
1- 1/4"-14 HHSD5 GRADE 5 CAN BE USED AS AN
ALTERNATE TO THE STAINLESS STEEL



Pedro De Figueiredo
PE 52609
09/16/2023

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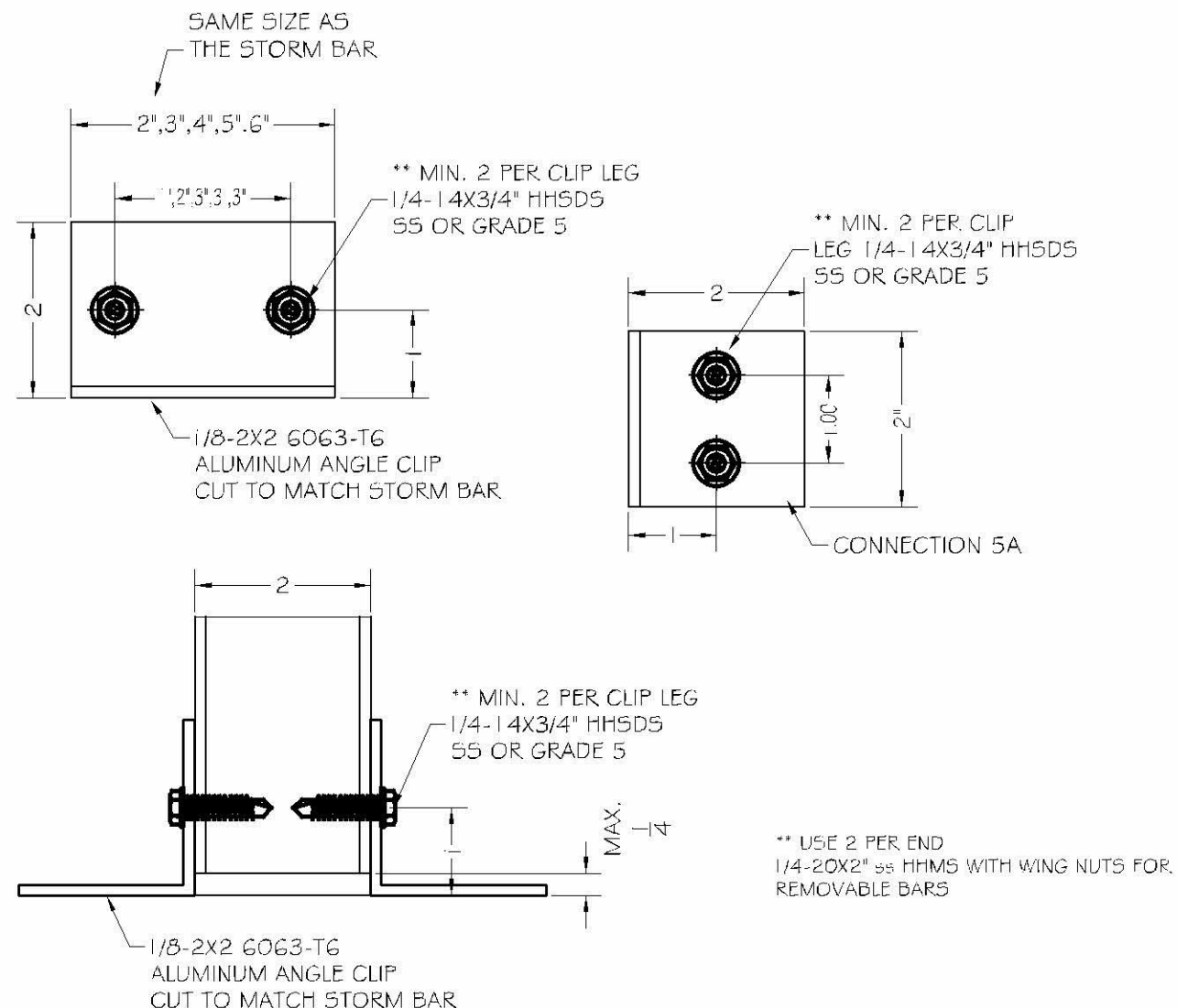


PRODUCT:
ES55 - ES44
EXTRUDED ALUMINUM ROLLING SHUTTER
LARGE & SMALL MISSILE
IMPACT RATED

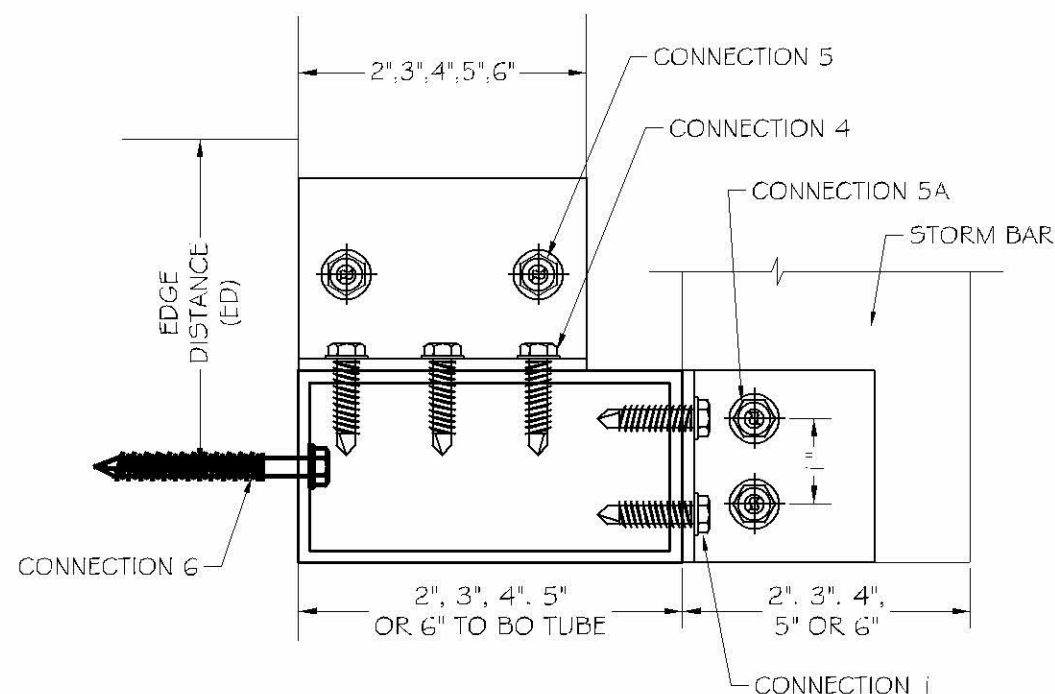
MANUFACTURER:
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11600 ADELMO LANE FORT MYERS, FL 33906
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SHEET
7 of 18

CONNECTION 5 CLIP ANCHORED TO STORM BAR



CONNECTION 6 BUILT-OUT WALL ANCHORAGE



CONNECTION 6 MINIMUM NUMBER OF ANCHORS TO SECURE BUILT-OUT TUBE TO WALL

STORM BARS	ANCHOR TYPE					
	A	B	C	D	E	G
2x2	4	5	4	3	3	3
2x3	5	7	5	4	5	5
2x4	7	8	7	4	6	6
2x5	8	10	8	5	7	7
2x6	9	11	9	6	8	8

CONNECTION 6A:
WHEN USED TO HOLD ONLY INSIDE STORM BAR WITH LOAD BEARING AGAINST THE WALL USE ONLY 2 ANCHORS TYPE A, B, C, D, E OR G

	ANCHOR TYPE					
	A	B	C	D	E	G
S	3.0	3.0	3.0	2.0	4.0	3.0
ED	2.5	2.5	1.0	0.5	4.0	2.5

S - ANCHOR SPACING
ED - EDGE DISTANCE

ANCHOR SCHEDULE

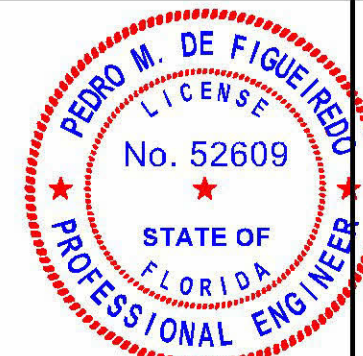
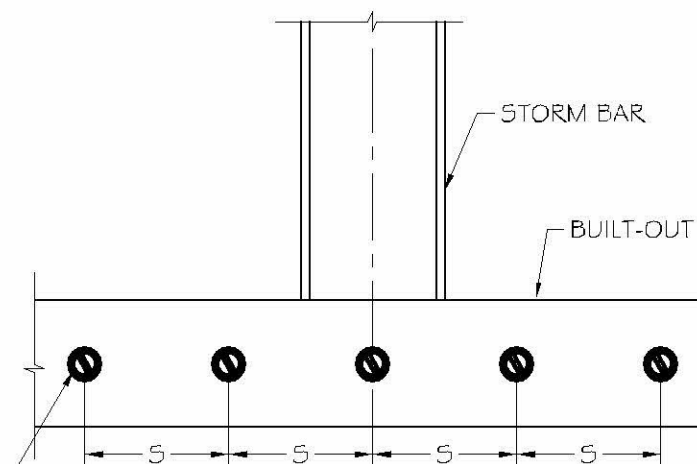
TYPE	DESCRIPTION	MANUFACTURER	EMBEDMENT	ED	BASE MATERIAL
A	1/4" SS TAPCON	ITW	1 3/4"	2 1/2"	3000 psi CONCRETE
B	1/4" SS TAPCON	ITW	1 1/4"	2 1/2"	HOLLOW BLOCK
C	1/4"-14 SS HHMS	GENERIC	1 1/2"	1"	WOOD (SG >= .46)
D	1/4"-14 SS HHSDS	GENERIC	1/8"	3/4"	1/8" METAL STRUCTURE
E	5/16" TAPCON	ITW	1 1/4"	4"	HOLLOW BLOCK
G	5/16" TAPCON	ITW	2 1/4"	2 1/2"	3000 psi CONCRETE

ANCHOR NOTES:

- 1- 1/4" TAPCONS CAN BE USED AS TYPE C INTO WOOD
- 2- 1/4" CARBON STEEL TAPCONS CAN BE USED AS AN ALTERNATE TO THE STAINLESS STEEL
- 3- METAL STRUCTURES ARE MADE OF MINIMUM 6063-T6 ALUMINUM OR ASTM A36 STEEL

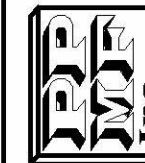
CONNECTION CAPACITY IS LIMITED TO THE MAXIMUM ALLOWABLE PRESSURES SPECIFIED AT TABLE 4 ON SHEET 5

QUANTITY OF ANCHORS FROM TABLE ARE TO BE EQUALLY DIVIDED FROM CENTER LINE OF STORM BAR



Pedro De Figueiredo
PE 52609
09/16/2023

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PRODUCT:
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EXTRUDED ALUMINUM ROLLING SHUTTER
LARGE & SMALL MISSILE
IMPACT RATED

MANUFACTURER:
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NORTH AMERICA
11600 ADELMO LANE FORT MYERS, FL 33966
TEL.: (239) 278-3066 - FAX: (239) 278-5977
EMAIL: INFO@CROCIUSA.COM

SCALE:
NA

DATE:
09/16/2023

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23-165

SHEET
8 of 18

ENGINEERING SERVICES
CA 82439

9715 W. BROWARD BLVD. 247

PLANTATION, FL 33324

PPMIF@AOL.COM

ENGINEER SEAL
PEDRO DE FIGUEIREDO
PE 52609

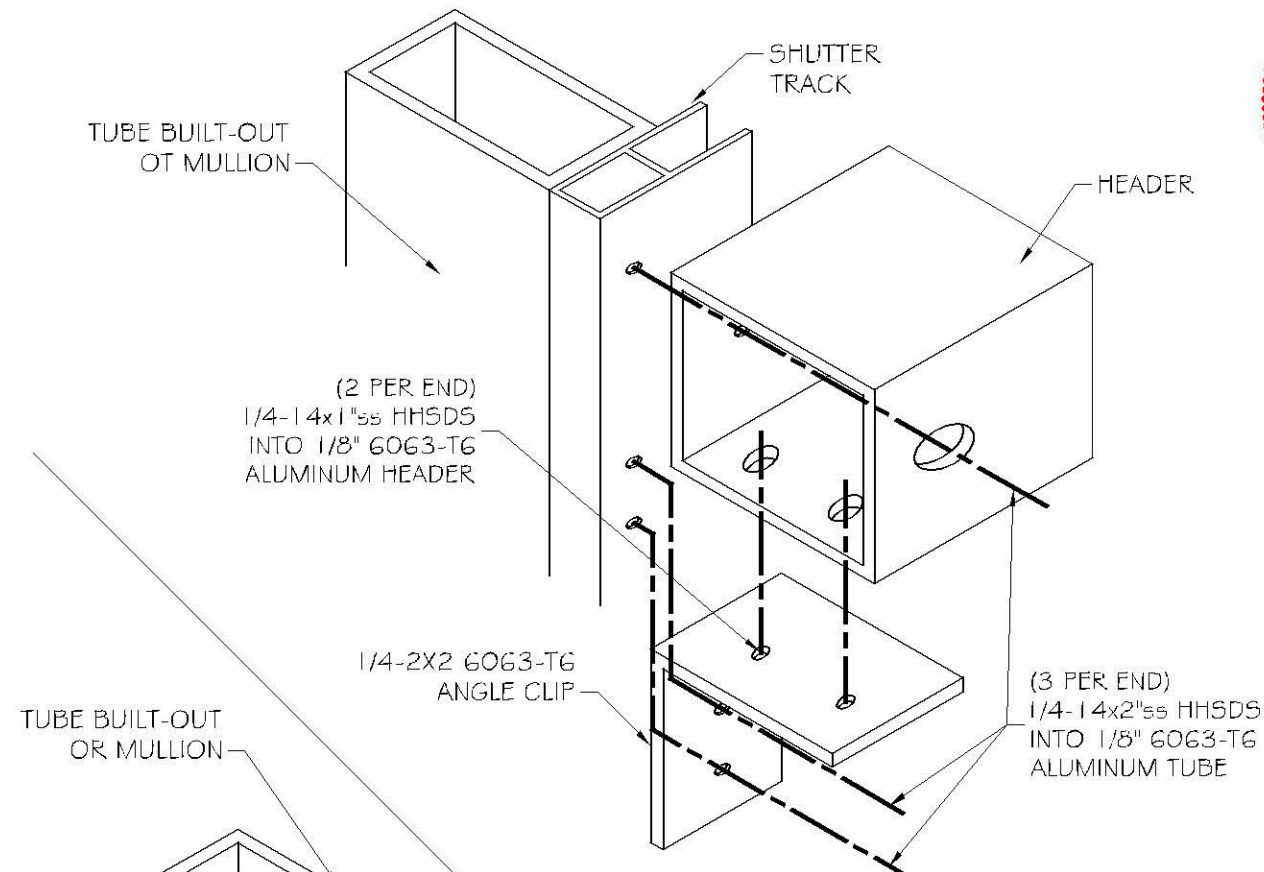
TABLE 5- HEADER BARS
CONNECTION TYPE 8
ALLOWABLE DESIGN LOAD - Pd (Psf)

TT	SBH	HEADER BAR				
		2X2	2X3	2X4	2X5	2X6
72	36	75	150	150	150	150
	48	57	136	147	147	147
	60	45	109	117	117	117
	72	38	91	98	98	98
	84	32	78	84	84	84
	96	28	68	73	73	73
	108	25	60	65	65	65
	120	23	54	59	59	59
84	36	48	127	150	150	150
	48	36	95	126	126	126
	60	29	76	100	100	100
	72	24	63	84	84	84
	84	20	54	72	72	72
	96	--	48	63	63	63
	108	--	42	56	56	56
	120	--	38	50	50	50
96	36	32	85	147	147	147
	48	24	64	110	110	110
	60	--	51	88	88	88
	72	--	43	73	73	73
	84	--	36	63	63	63
	96	--	32	55	55	55
	108	--	28	49	49	49
	120	--	26	44	44	44
108	36	22	60	121	130	130
	48	--	45	91	98	98
	60	--	36	73	78	78
	72	--	30	61	65	65
	84	--	26	52	56	56
	96	--	22	45	49	49
	108	--	20	40	43	43
	120	--	--	36	39	39
120	36	--	44	88	117	117
	48	--	33	66	88	88
	60	--	26	53	70	70
	72	--	22	44	59	59
	84	--	--	38	50	50
	96	--	--	33	44	44
	108	--	--	29	39	39
	120	--	--	26	35	35
130	36	--	34	69	108	108
	48	--	26	52	81	81
	60	--	21	42	65	65
	72	--	--	35	54	54
	84	--	--	30	46	46
	96	--	--	26	41	41
	108	--	--	23	36	36
	120	--	--	21	32	32
150	36	--	22	45	79	94
	48	--	--	34	59	70
	60	--	--	27	47	56
	72	--	--	23	39	47
	84	--	--	--	34	40
	96	--	--	--	30	35
	108	--	--	--	26	31
	120	--	--	--	24	28

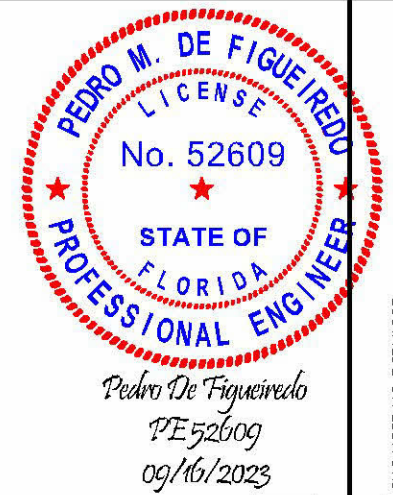
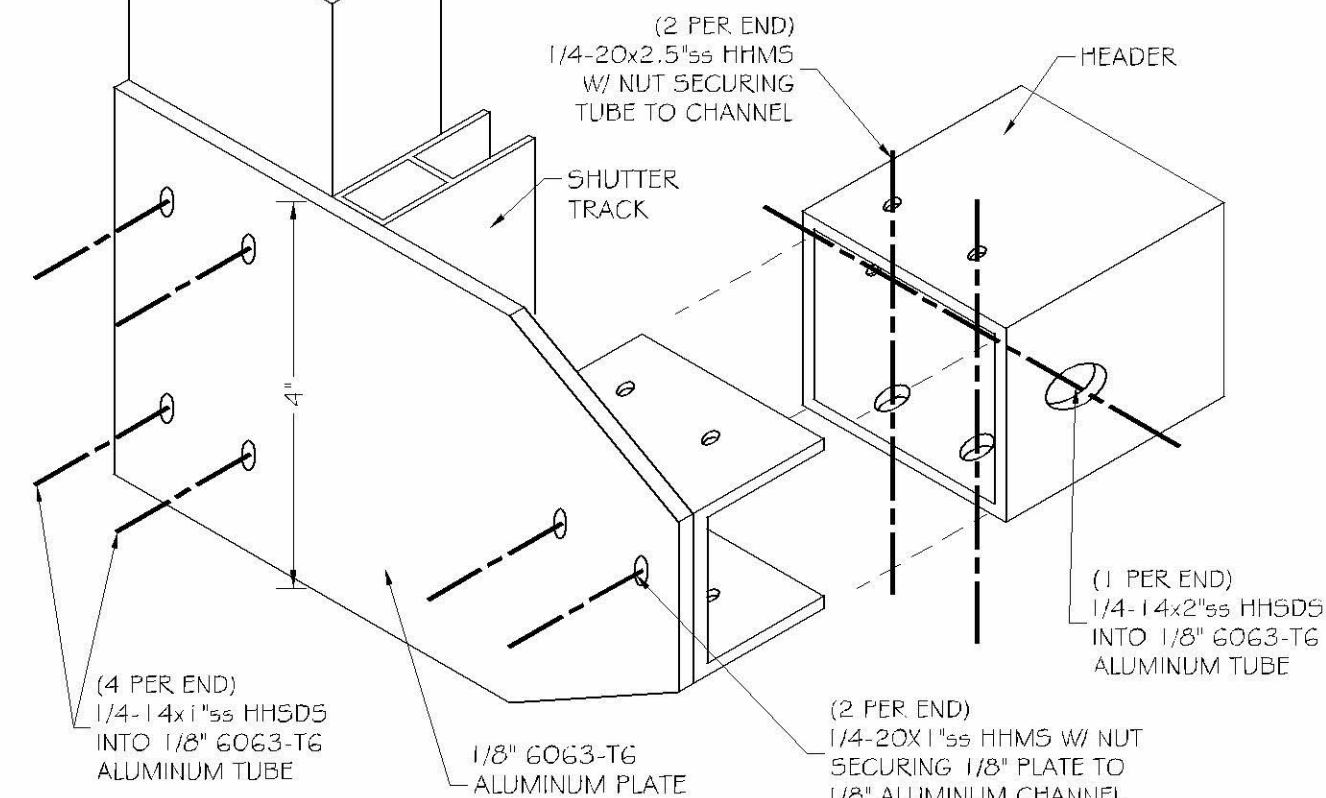
TABLE 6- HEADER BARS
CONNECTION TYPE 9
ALLOWABLE DESIGN LOAD - Pd (Psf)

TT	SBH	HEADER BAR TYPE				
		2X2	2X3	2X4	2X5	2X6
72	36	75	150	150	150	150
	48	57	136	150	150	150
	60	45	109	150	150	150
	72	38	91	138	150	150
	84	32	78	118	147	147
	96	28	68	103	129	129
	108	25	60	92	115	115
	120	23	54	83	103	103
84	36	48	127	150	150	150
	48	36	95	150	150	150
	60	29	76	122	150	150
	72	24	63	101	141	147
	84	20	54	87	121	126
	96	--	48	76	106	111
	108	--	42	68	94	98
	120	--	38	61	85	88
96	36	32	85	150	150	150
	48	24	64	116	150	150
	60	--	51	93	130	150
	72	--	43	78	108	129
	84	--	36	67	93	111
	96	--	32	58	81	97
	108	--	28	52	72	86
	120	--	26	47	65	77
108	36	22	60	121	150	150
	48	--	45	91	128	150
	60	--	36	73	103	136
	72	--	30	61	86	114
	84	--	26	52	73	97
	96	--	22	45	64	85
	108	--	20	40	57	76
	120	--	--	36	51	68
120	36	--	44	88	139	150
	48	--	33	66	104	138
	60	--	26	53	83	110
	72	--	22	44	69	92
	84	--	--	38	59	79
	96	--	--	33	52	69
	108	--	--	29	46	61
	120	--	--	26	42	55
130	36	--	34	69	118	150
	48	--	26	52	89	118
	60	--	21	42	71	94
	72	--	--	35	59	78
	84	--	--	30	51	67
	96	--	--	26	44	59
	108	--	--	23	39	52
	120	--	--	21	35	47
150	36	--	22	45	79	118
	48	--	--	34	59	88
	60	--	--	27	47	71
	72	--	--	23	39	59
	84	--	--	--	34	50
	96	--	--	--	30	44
	108	--	--	--	26	39
	120	--	--	--	24	35

HEADER MOUNT - CONNECTION 8



HEADER MOUNT - CONNECTION 9



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CA 92439
9715 W. BROWARD BLVD. 247
PLANTATION, FL 33324
PPMIF@AOL.COM



PRODUCT:
E555 - ES44
EXTRUDED ALUMINUM ROLLING SHUTTER
LARGE & SMALL MISSILE
IMPACT RATED

MANUFACTURER:
CROCI
NORTH AMERICA
11600 ADELMO LANE FORT MYERS, FL 33966
TEL.: (239) 278-3066 - FAX: (239) 278-5977
EMAIL: INFO@CROCIUSA.COM

SCALE:

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DATE:

09/16/2023

DRAWING #

23-165

SHEET

9 of 18

HEADER NOTES:

1- TT: TRACK TO TRACK, SBH: STORM BAR HEIGHT

2- SPECIFIC CONDITIONS OUT OF THE SCOPE OF THESE TABLES CAN BE ADDRESSED WITH A SITE SPECIFIC DESIGN BY A PROFESSIONAL ENGINEER.

DIVIDE EQUALLY FROM CENTER LINE OF HEADER
THE MINIMUM QUANTITY OF ANCHORS SELECTED
FROM TABLES BELOW

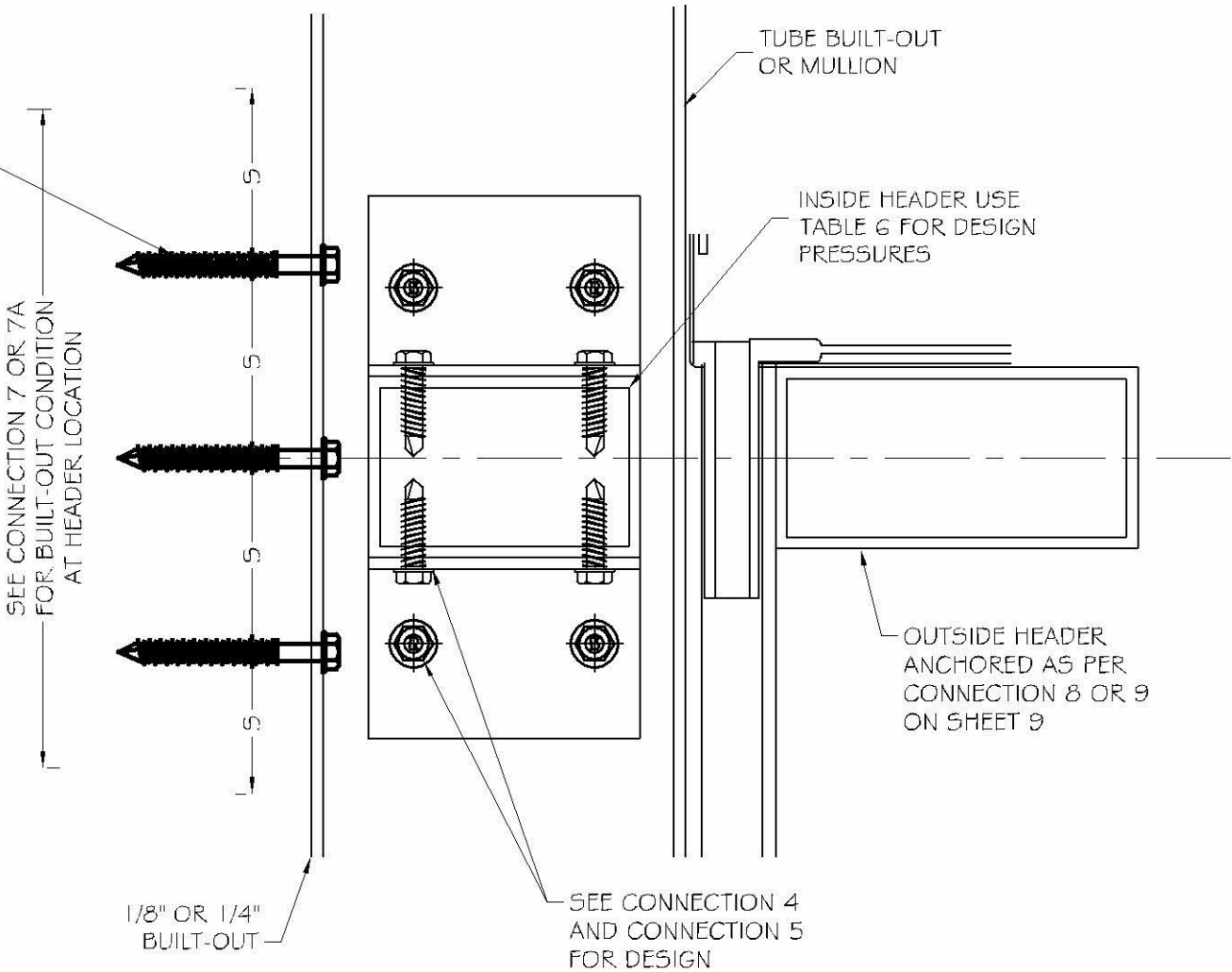
CONNECTION 7
MINIMUM NUMBER OF ANCHORS
TO SECURE 1/8" BUILT-OUT TUBE TO WALL

HEADER TYPE	CONNECTION 8						CONNECTION 9					
	A	B	C	D	E	G	A	B	C	D	E	G
2X2	1	2	1	1	1	1	2	3	2	2	2	2
2X3	3	4	3	2	3	3	5	6	5	3	4	4
2X4	5	6	5	3	4	4	-	-	-	4	6	6
2X5	5	6	5	3	4	4	-	-	-	5	-	-
2X6	5	6	5	3	4	4	-	-	-	5	-	-

CONNECTION 7A
MINIMUM NUMBER OF ANCHORS
TO SECURE 1/4" BUILT-OUT TUBE TO WALL

HEADER TYPE	CONNECTION 8						CONNECTION 9					
	A	B	C	D	E	G	A	B	C	D	E	G
2X2	1	3	1	2	2	1	1	3	1	2	2	1
2X3	2	6	2	3	4	2	2	6	2	3	4	2
2X4	3	6	3	3	4	2	4	--	4	4	5	3
2X5	3	6	3	3	4	2	4	--	4	5	6	4
2X6	3	6	3	3	4	2	4	--	4	5	6	4

S SPACING	ANCHOR TYPE					
	A	B	C	D	E	G
	3	3	2	2	4	3

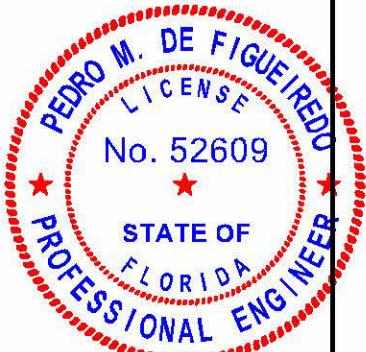


ANCHOR SCHEDULE

TYPE	DESCRIPTION	MANUFACTURER	EMBEDMENT	ED	BASE MATERIAL
A	1/4" 55 TAPCON	ITW	1 3/4"	2 1/2"	3000 psi CONCRETE
B	1/4" 55 TAPCON	ITW	1 1/4"	2 1/2"	HOLLOW BLOCK
C	1/4"-14 55 HHSM5	GENERIC	1 1/2"	1"	WOOD (SG > .46)
D	1/4"-14 55 HHSD5	GENERIC	1/8"	3/4"	1/8" METAL STRUCTURE
E	5/16" TAPCON	ITW	1 1/4"	4"	HOLLOW BLOCK
G	5/16" TAPCON	ITW	2 1/4"	2 1/2"	3000 psi CONCRETE

ANCHOR NOTES:

- 1- 1/4" TAPCONS CAN BE USED AS TYPE C INTO WOOD
- 2- 1/4" CARBON STEEL TAPCONS CAN BE USED AS AN ALTERNATE TO THE STAINLESS STEEL
- 3- METAL STRUCTURES ARE MADE OF MINIMUM 6063-T6 ALUMINUM OR ASTM A36 STEEL
- 4- SPECIFIC CONDITIONS OUT OF THE SCOPE OF THESE TABLES CAN BE ADDRESSED WITH A SITE SPECIFIC DESIGN BY A PROFESSIONAL ENGINEER.



Pedro De Figueiredo
PE 52609
09/16/2023

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PRODUCT:
E555 - E544
EXTRUDED ALUMINUM ROLLING SHUTTER
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IMPACT RATED

MANUFACTURER:
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11600 ADELMO LANE FORT MYERS, FL 33966
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10 of 18

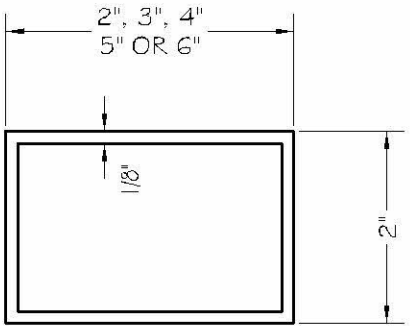
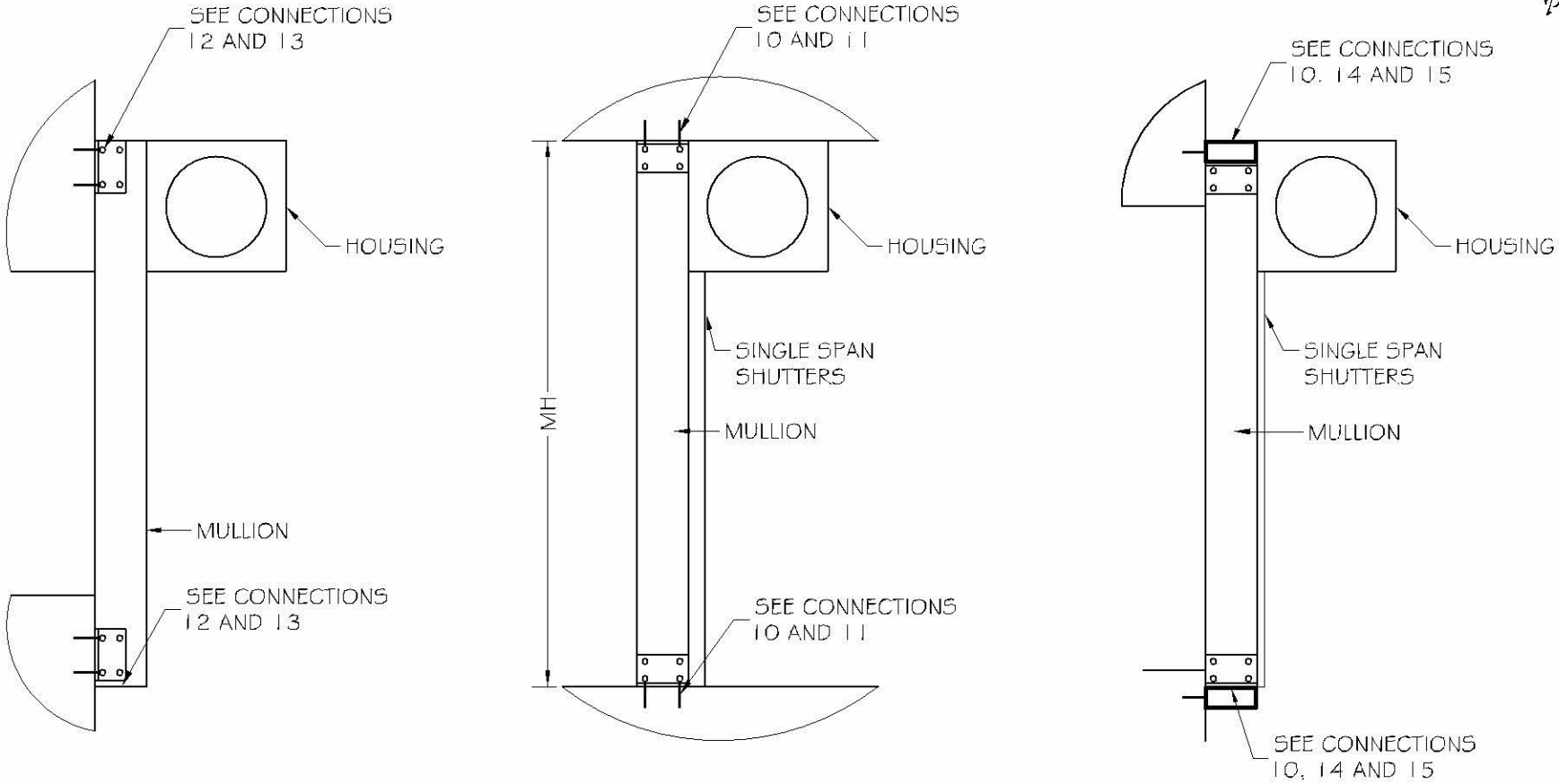
TABLE 7- MULLION BARS
ALLOWABLE DESIGN LOAD - Pd (Psf)

SS	MH	MULLION BAR TYPE				
		2X2	2X3	2X4	2X5	2X6
24	48	150	150	150	150	150
	60	150	150	150	150	150
	72	114	150	150	150	150
	84	72	150	150	150	150
	96	48	128	150	150	150
	108	33	90	150	150	150
	120	24	65	133	150	150
	132	--	44	91	150	150
30	48	150	150	150	150	150
	60	150	150	150	150	150
	72	91	150	150	150	150
	84	57	150	150	150	150
	96	38	102	150	150	150
	108	27	72	143	150	150
	120	--	52	106	150	150
	132	--	35	73	127	150
36	48	150	150	150	150	150
	60	132	150	150	150	150
	72	76	150	150	150	150
	84	48	128	150	150	150
	96	32	85	150	150	150
	108	22	60	119	150	150
	120	--	43	89	133	150
	132	--	29	60	106	144
48	48	100	100	110	110	110
	60	99	100	110	110	110
	72	57	100	110	110	110
	84	36	96	110	110	110
	96	24	64	110	110	110
	108	--	45	89	110	110
	120	--	32	66	100	110
	132	--	22	45	79	108
54	48	100	100	100	100	100
	60	88	100	100	100	100
	72	50	100	100	100	100
	84	32	85	100	100	100
	96	21	57	100	100	100
	108	--	40	79	100	100
	120	--	29	59	89	100
	132	--	--	40	70	100
60	48	85	85	85	85	85
	60	79	85	85	85	85
	72	45	85	85	85	85
	84	28	76	85	85	85
	96	--	51	85	85	85
	108	--	36	71	80	85
	120	--	26	53	63	85
	132	--	--	36	85	85
80	48	40	40	40	40	40
	60	40	40	40	40	40
	72	34	40	40	40	40
	84	21	40	40	40	40
	96	--	38	40	40	40
	108	--	27	40	40	40
	120	--	--	40	40	40
	132	--	--	27	40	40

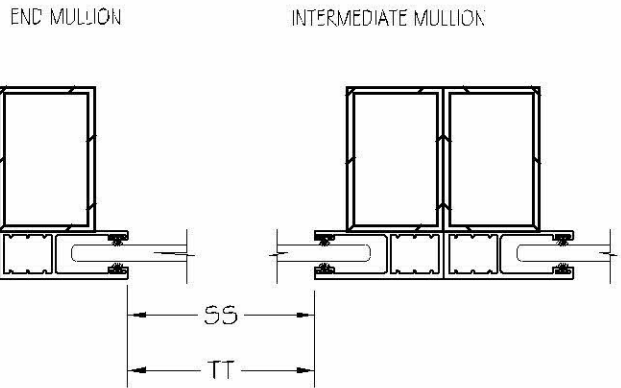
STORM BAR NOTES:
1- SS: SLAT SPAN, MH: MULLION BAR HEIGHT
2- TABLES MUST WORK IN CONJUNCTION WITH MAXIMUM ALLOWABLE SLAT SPAN TABLES ON SHEET 2.
3- THESE TABLES ARE FURTHER LIMITED BY TYPE OF ANCHORAGE USED.
4- SPECIFIC CONDITIONS OUT OF THE SCOPE OF THESE TABLES CAN BE ADDRESSED WITH A SITE SPECIFIC DESIGN BY A PROFESSIONAL ENGINEER.

MULLION SELECTION TABLES
ALLOWABLE DESIGN LOAD - Pd (Psf)

CASE 1 :
SINGLE SPAN SHUTTERS



SHAPES:
1/8-2X2 6063-T6
1/8-2X3 6063-T6
1/8-2X4 6063-T6
1/8-2X5 6063-T6
1/8-2X6 6063-T6



Pedro De Figueiredo
PE 52609
09/16/2023

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NORTH AMERICA
11600 ADELMO LANE FORT MYERS, FL 33966
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11 of 18

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ENGINEER SEAL
PEDRO DE FIGUEIREDO
PE 52609

TABLE 8- MULLION BARS
ALLOWABLE DESIGN LOAD - Pd (Psf)

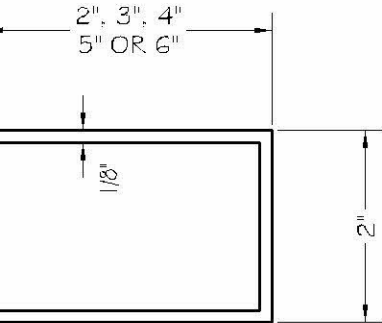
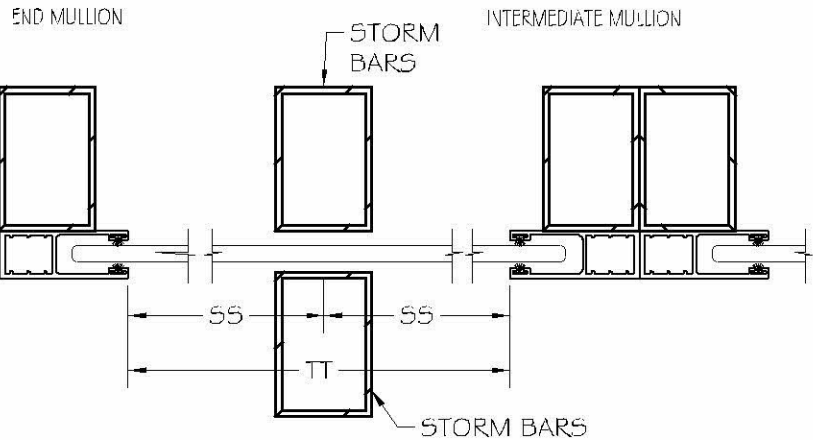
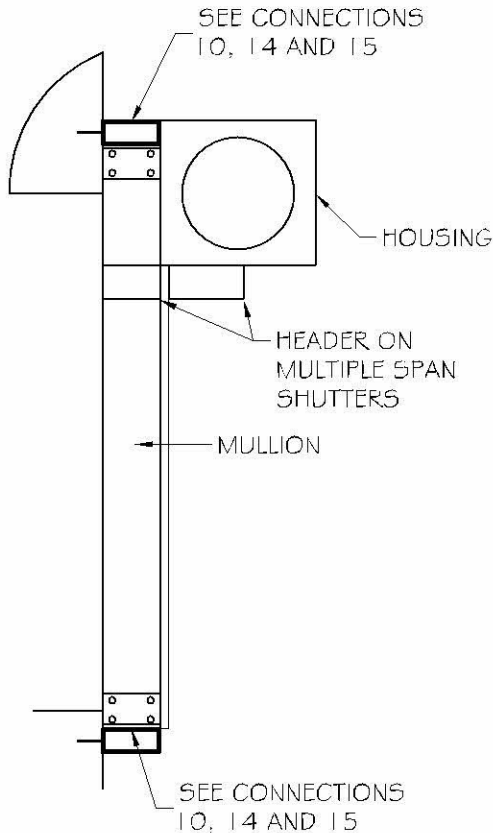
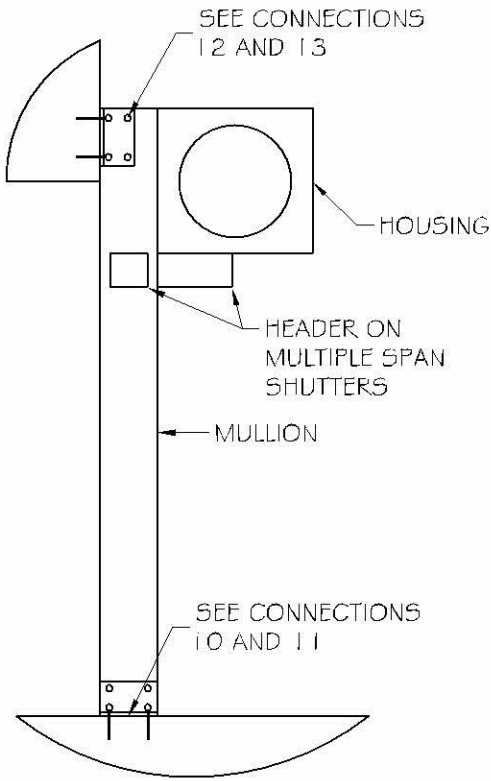
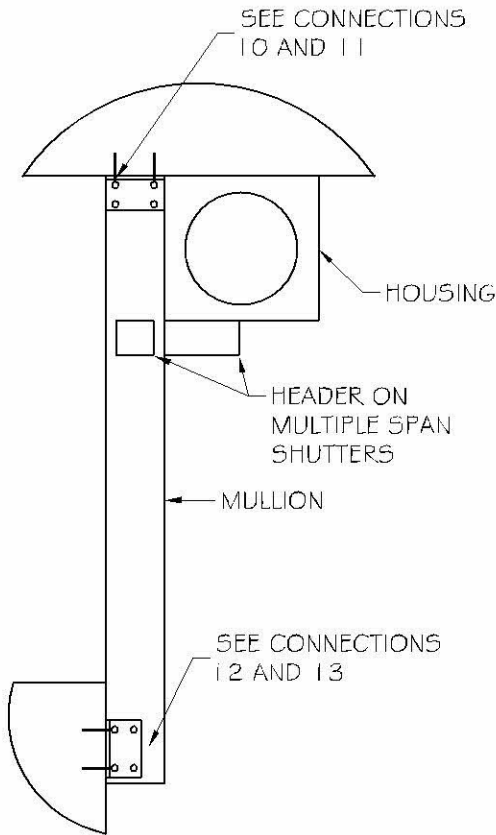
TT	MH	MULLION BAR TYPE				
		2X2	2X3	2X4	2X5	2X6
72	48	75	150	150	150	150
	60	57	120	150	150	150
	72	45	95	150	150	150
	84	29	80	128	150	150
	96	--	52	100	135	135
	108	--	34	78	115	118
	120	--	23	58	92	104
	132	--	--	36	75	93
84	48	48	127	150	150	150
	60	36	95	150	150	150
	72	29	76	122	150	150
	84	21	63	101	137	137
	96	--	54	87	116	116
	108	--	39	76	101	101
	120	--	27	62	89	89
	132	--	--	39	76	80
96	48	32	85	150	150	150
	60	24	64	116	150	150
	72	--	51	93	130	142
	84	--	43	78	108	120
	96	--	36	67	93	102
	108	--	32	58	81	88
	120	--	28	52	72	78
	132	--	--	42	65	70
108	48	22	60	121	150	150
	60	--	45	91	128	150
	72	--	36	73	103	126
	84	--	30	61	86	106
	96	--	26	52	73	90
	108	--	22	45	64	78
	120	--	20	40	57	69
	132	--	--	36	51	62
120	48	--	44	88	139	150
	60	--	33	66	104	138
	72	--	26	53	83	110
	84	--	22	44	69	92
	96	--	--	38	59	79
	108	--	--	33	52	69
	120	--	--	29	46	61
	132	--	--	26	42	55
130	48	--	34	69	118	150
	60	--	26	52	89	118
	72	--	21	42	71	94
	84	--	--	35	59	78
	96	--	--	30	51	67
	108	--	--	26	44	59
	120	--	--	23	39	52
	132	--	--	21	35	47
150	48	--	22	45	79	118
	60	--	--	34	59	88
	72	--	--	27	47	71
	84	--	--	23	39	59
	96	--	--	--	34	50
	108	--	--	--	30	44
	120	--	--	--	26	39
	132	--	--	--	24	35

STORM BAR NOTES:
1- TT: TRACK TO TRACK, MH: MULLION BAR HEIGHT
2- TABLES MUST WORK IN CONJUNCTION WITH MAXIMUM ALLOWABLE SLAT SPAN TABLES ON SHEET 2.
3- THESE TABLES ARE FURTHER LIMITED BY TYPE OF ANCHORAGE USED.
4- SPECIFIC CONDITIONS OUT OF THE SCOPE OF THESE TABLES CAN BE ADDRESSED WITH A SITE SPECIFIC DESIGN BY A PROFESSIONAL ENGINEER.

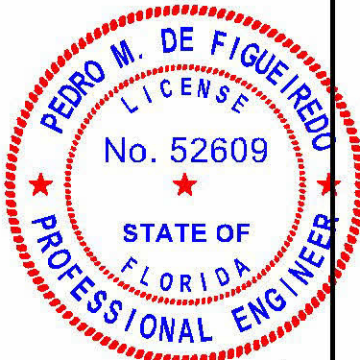
MULLION SELECTION TABLES

ALLOWABLE DESIGN LOAD - Pd (Psf)

CASE 2:
MULTIPLE SPAN SHUTTERS WITH
HEADER ANCHORED TO MULLION



SHAPES:
1/8-2X2 6063-T6
1/8-2X3 6063-T6
1/8-2X4 6063-T6
1/8-2X5 6063-T6
1/8-2X6 6063-T6



Pedro De Figueiredo
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PRODUCT:
E555 - E544
EXTRUDED ALUMINUM ROLLING SHUTTER
LARGE & SMALL MISSILE
IMPACT RATED

MANUFACTURER:
CROCI
NORTH AMERICA
11600 ADELMO LANE FORT MYERS, FL 33966
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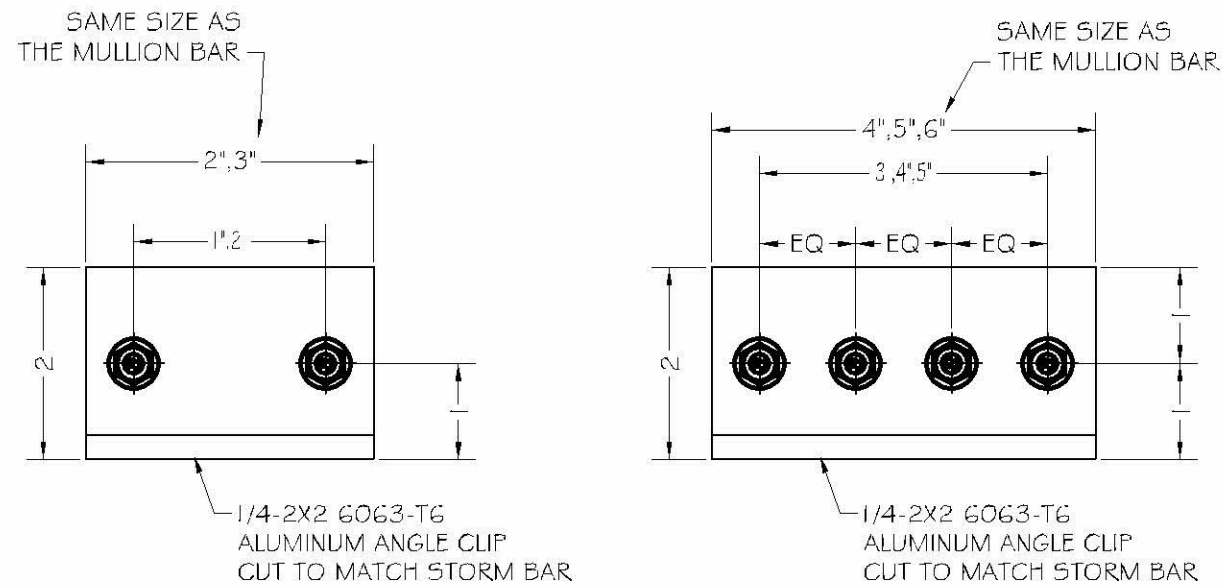
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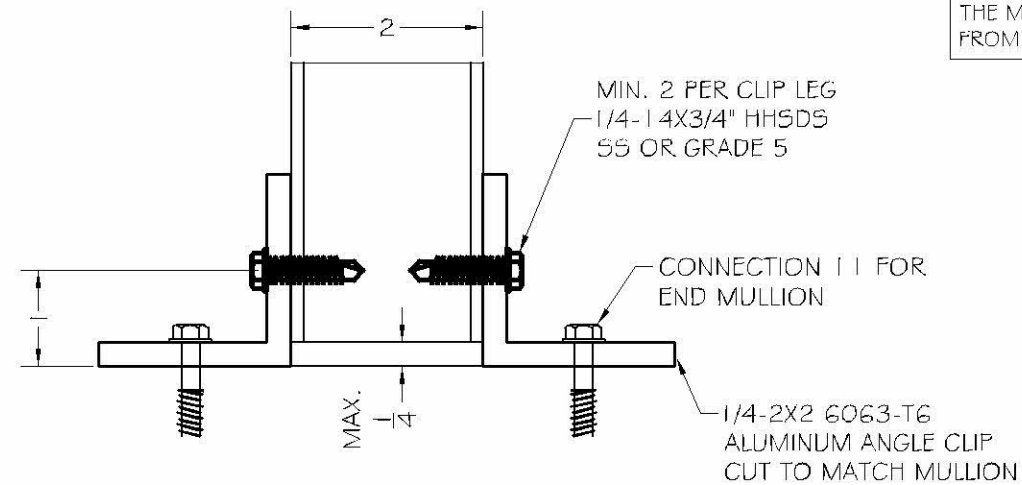
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23-165

SHEET
12 of 18

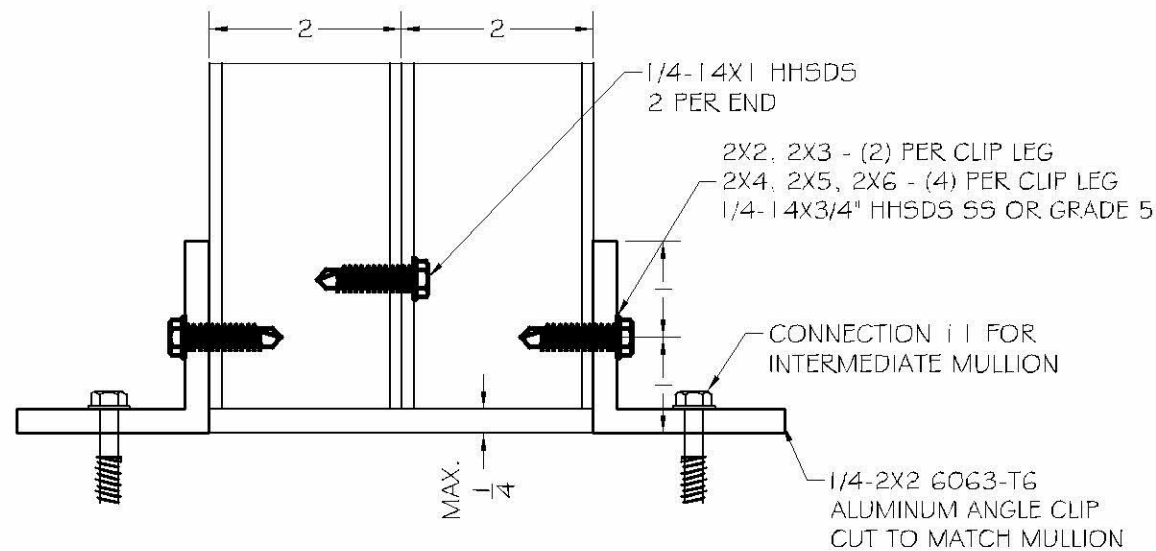
CONNECTION 10 CLIP ANCHORED TO MULLION



END MULL

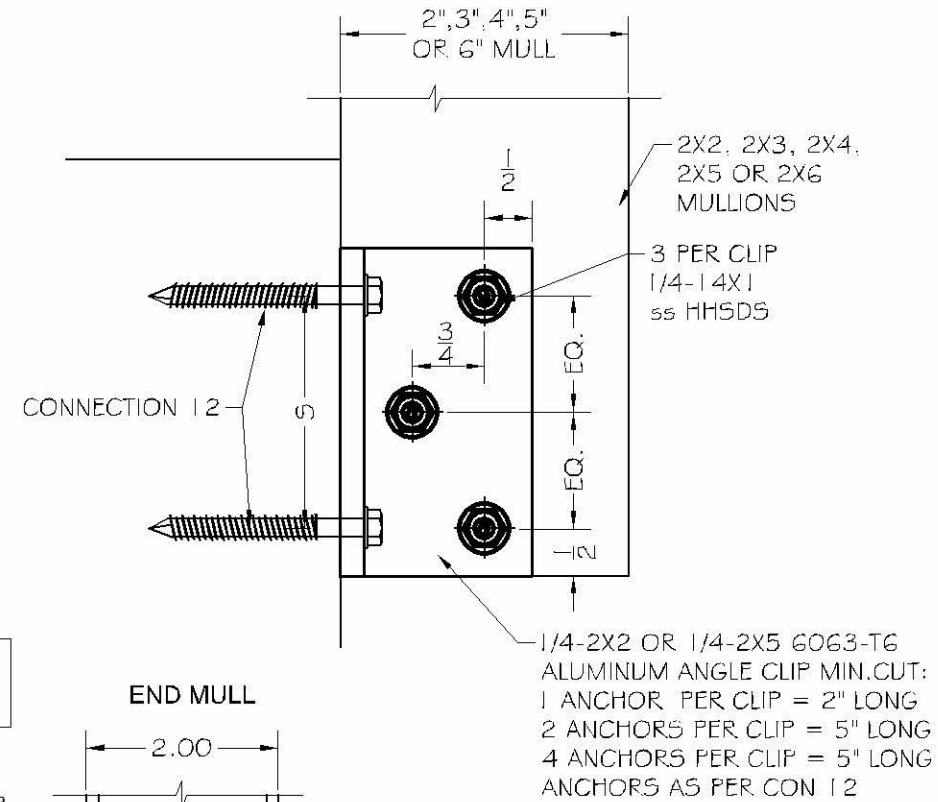


INTERMEDIATE MULL



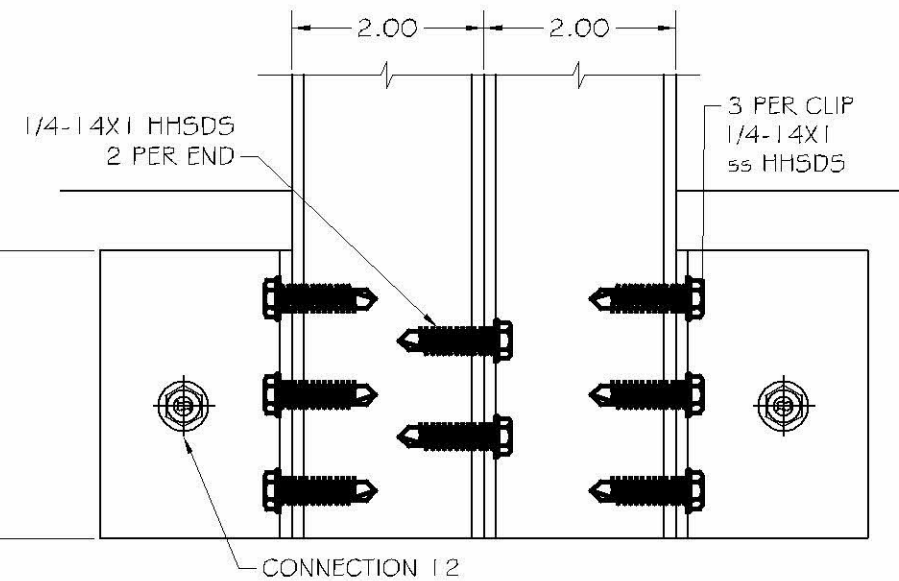
ANCHORS TO BE DESIGNED BASED ON THE MINIMUM MULLION SIZE SELECTED FROM TABLES 7 AND 8.

CONNECTION 13 CLIP ANCHORED TO MULLION BAR



1/4-2X2 OR 1/4-2X5 6063-T6 ALUMINUM ANGLE CLIP MIN. CUT:
1 ANCHOR PER CLIP = 2" LONG
2 ANCHORS PER CLIP = 5" LONG
4 ANCHORS PER CLIP = 5" LONG
ANCHORS AS PER CON 12

INTERMEDIATE MULL



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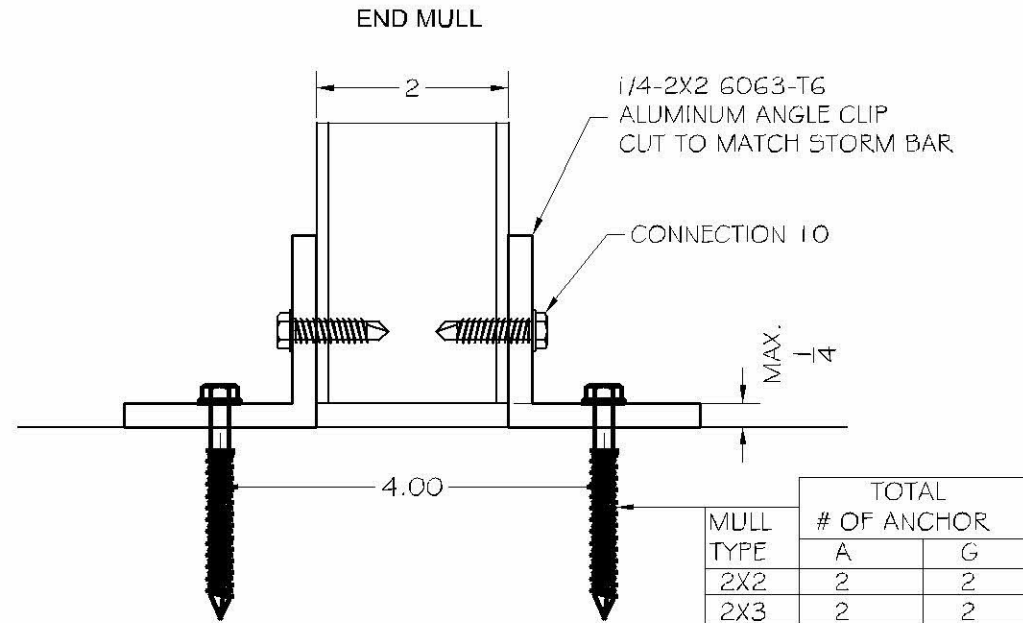


PRODUCT:
ES55 - ES44
EXTRUDED ALUMINUM ROLLING SHUTTER
LARGE & SMALL MISSILE
IMPACT RATED

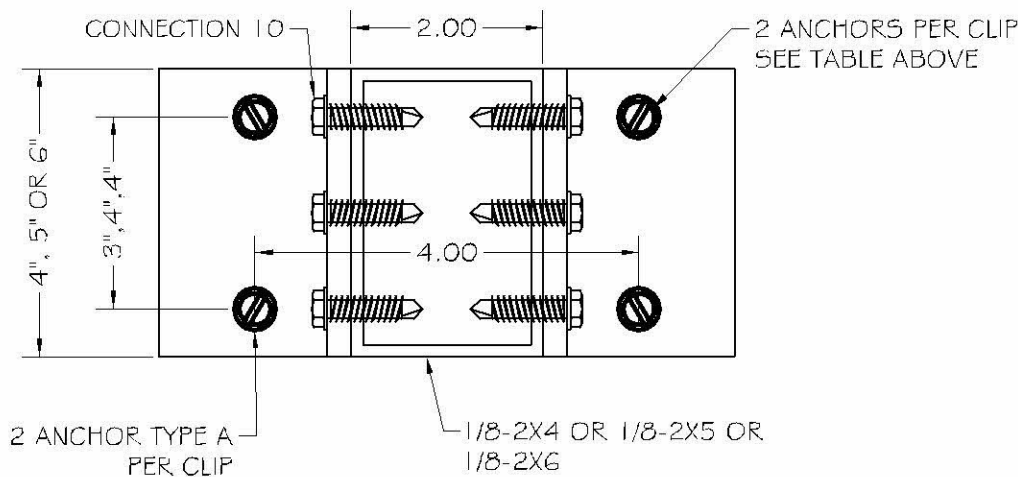
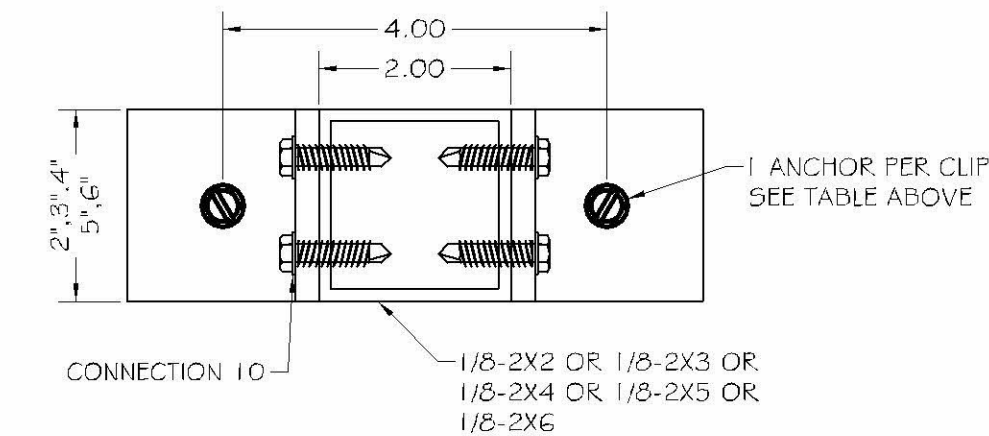
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13 of 18

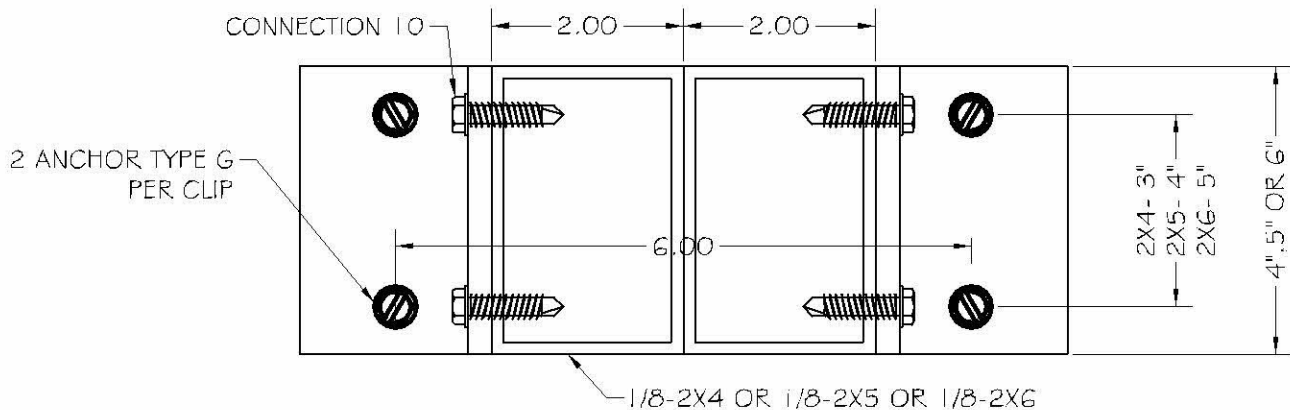
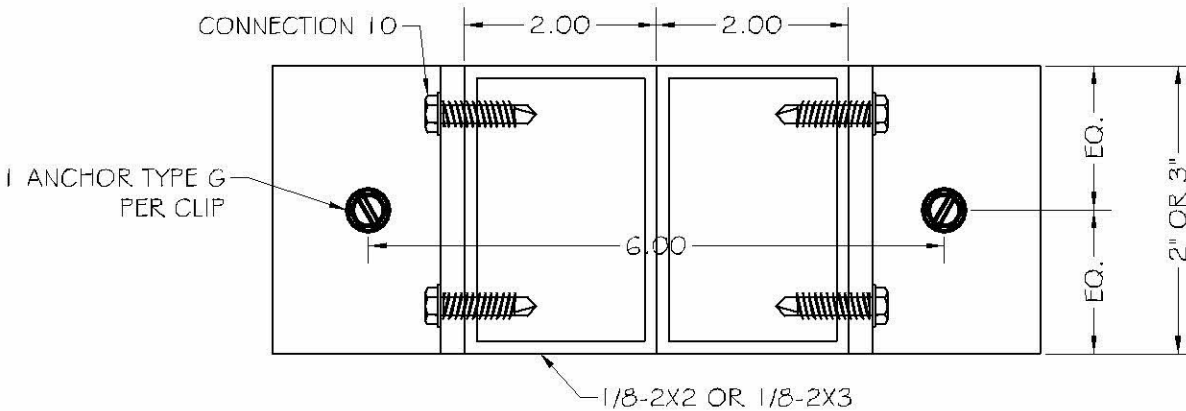
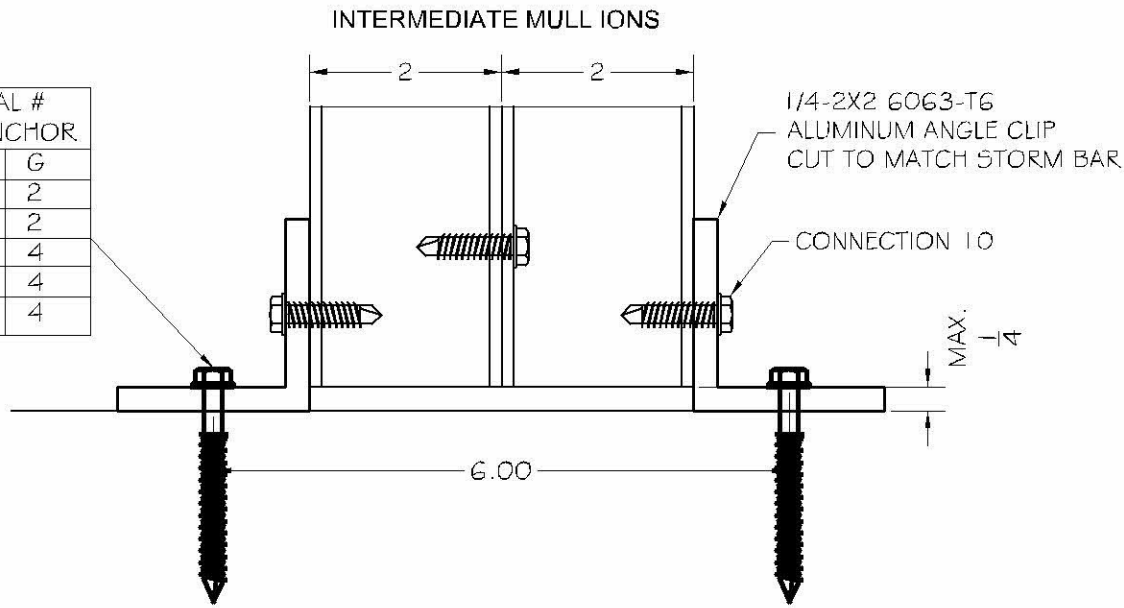
CONNECTION 11
CLIP ANCHORED TO CONCRETE FLOOR/CEILING



ANCHORS TO BE DESIGNED BASED ON THE MINIMUM MULLION SIZE SELECTED FROM TABLES 7 AND 8.



MULL TYPE	TOTAL # OF ANCHOR	
	A	G
2X2	NA	2
2X3	NA	2
2X4	NA	4
2X5	NA	4
2X6	NA	4



ANCHOR SCHEDULE

TYPE	DESCRIPTION	MANUFACTURER	EMBEDMENT	ED	BASE MATERIAL
A	1/4" ss TAPCON	ITW	1 3/4"	2 1/2"	3000 psi CONCRETE
G	5/16" TAPCON	TW	2 1/4"	2 1/2"	3000 psi CONCRETE

ANCHOR NOTES:

1- 1/4" CARBON STEEL TAPCONS CAN BE USED AS AN ALTERNATE TO THE STAINLESS STEEL



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PRODUCT:
E544
E555 - ES44
EXTRUDED ALUMINUM ROLLING SHUTTER
LARGE & SMALL MISSILE
IMPACT RATED

MANUFACTURER:
CROCI
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SCALE:

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DATE:

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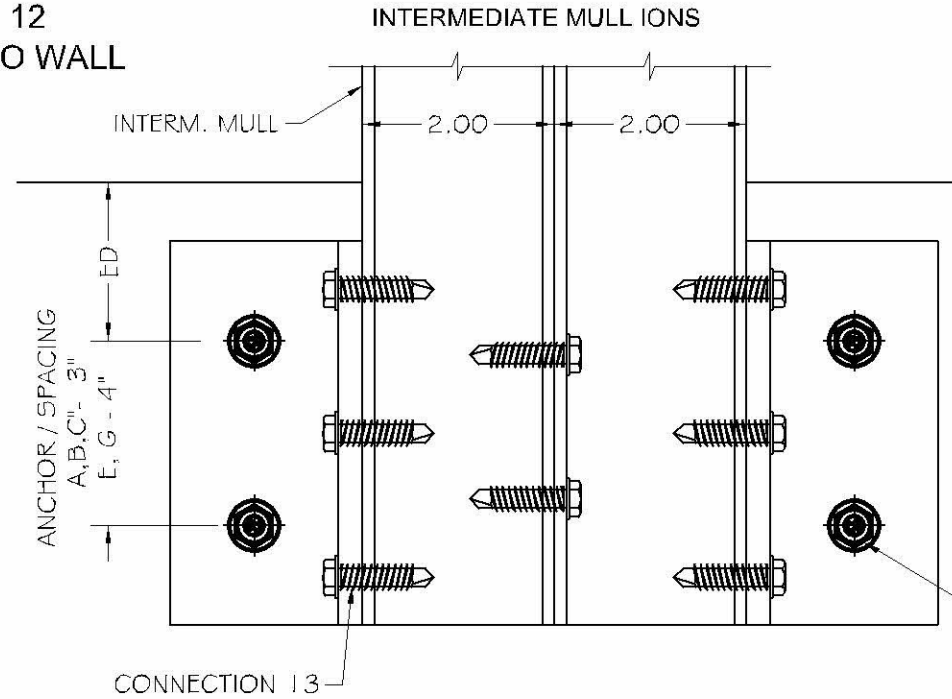
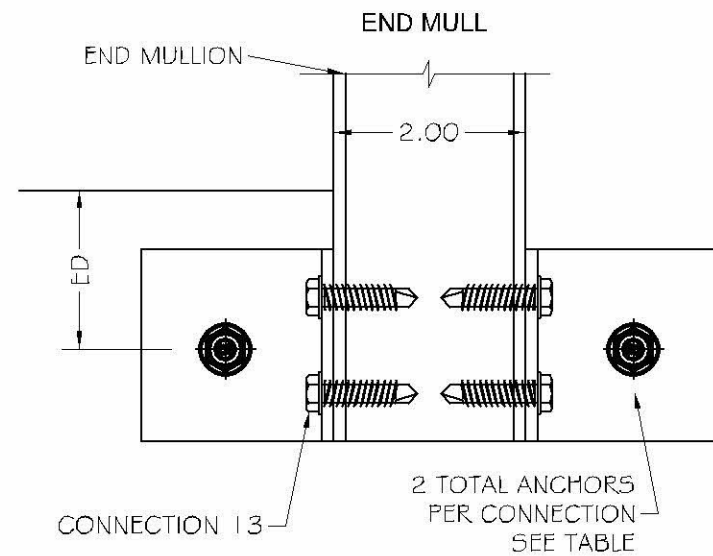
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23-165

SHEET

14 of 18

CONNECTION 12
CLIP ANCHORED TO WALL

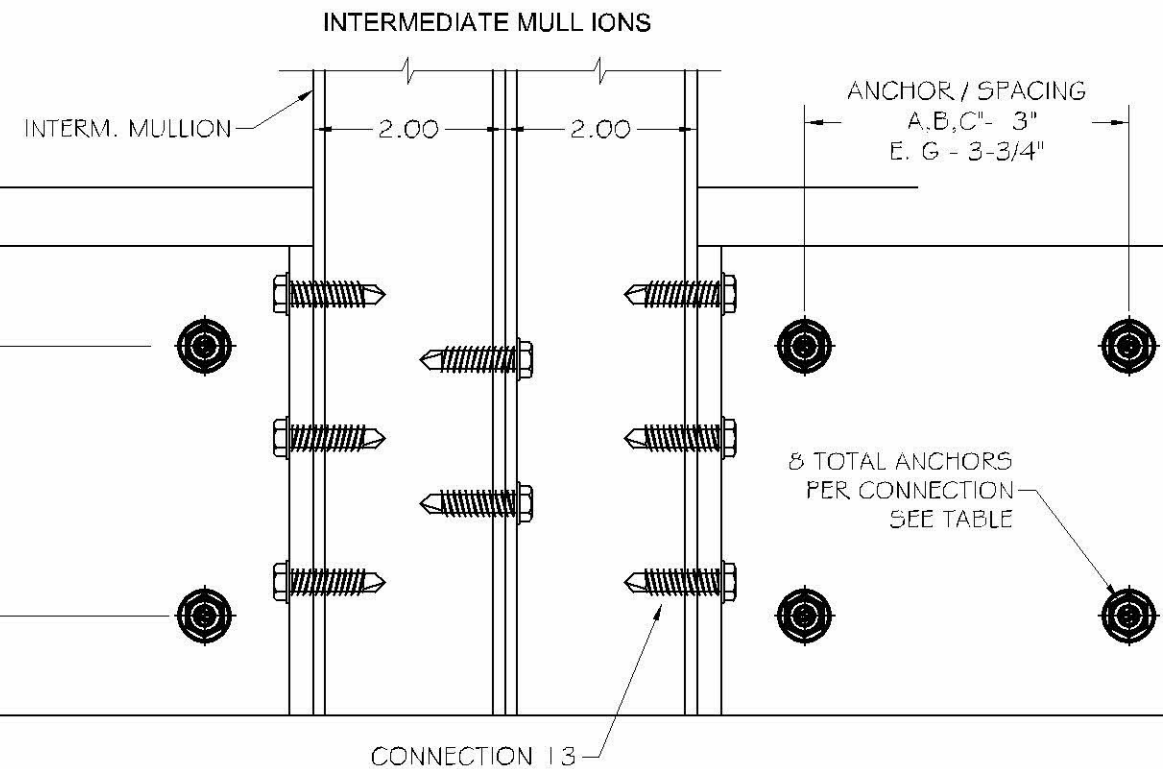


INTERMEDIATE MULL					
MULL TYPE	TOTAL # OF ANCHOR				
	A	B	C	G	E
2X2	4	8	4	4	4
2X3	4	-	4	4	8
2X4	8	-	8	8	-
2X5	8	-	8	8	-
2X6	8	-	8	8	-

4 TOTAL ANCHORS
PER CONNECTION
SEE TABLE

END MULL					
MULL TYPE	TOTAL # OF ANCHOR				
	A	B	C	G	E
2X2	2	4	2	2	2
2X3	2	-	2	2	4
2X4	4	-	4	4	-
2X5	4	-	4	4	-
2X6	4	-	4	4	-

ANCHORS TO BE DESIGNED BASED ON
THE MINIMUM MULLION SIZE SELECTED
FROM TABLES 7 AND 8.



ANCHOR SCHEDULE

TYPE	DESCRIPTION	MANUFACTURER	EMBEDMENT	ED	BASE MATERIAL
A	1/4" 304 SS TAPCON	ITW	1 3/4"	2 1/2"	3000 psi CONCRETE
B	1/4" 304 SS TAPCON	ITW	1 1/4"	2 1/2"	HOLLOW BLOCK
C	1/4"-14 304 SS HHSM	GENERIC	1 1/2"	1"	WOOD (SG >= .46)
E	5/16" TAPCON	ITW	1 1/4"	4"	HOLLOW BLOCK
G	5/16" TAPCON	ITW	2 1/4"	2 1/2"	3000 psi CONCRETE

ANCHOR NOTES:

- 1- 1/4" TAPCONS CAN BE USED AS TYPE C INTO WOOD
- 2- 1/4" CARBON STEEL TAPCONS CAN BE USED AS AN ALTERNATE TO THE STAINLESS STEEL
- 3- METAL STRUCTURES ARE MADE OF MINIMUM 6063-T6 ALUMINUM OR ASTM A36 STEEL



Pedro De Figueiredo
PE 52609
09/16/2023

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PRODUCT:
ES55 - ES44
EXTRUDED ALUMINUM ROLLING SHUTTER
LARGE & SMALL MISSILE
IMPACT RATED

MANUFACTURER:
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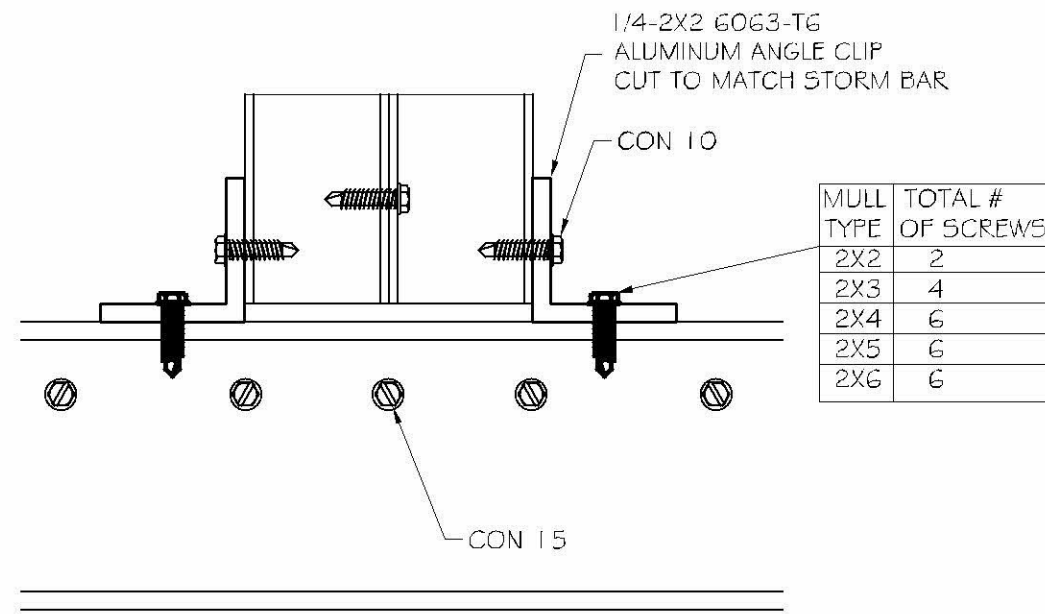
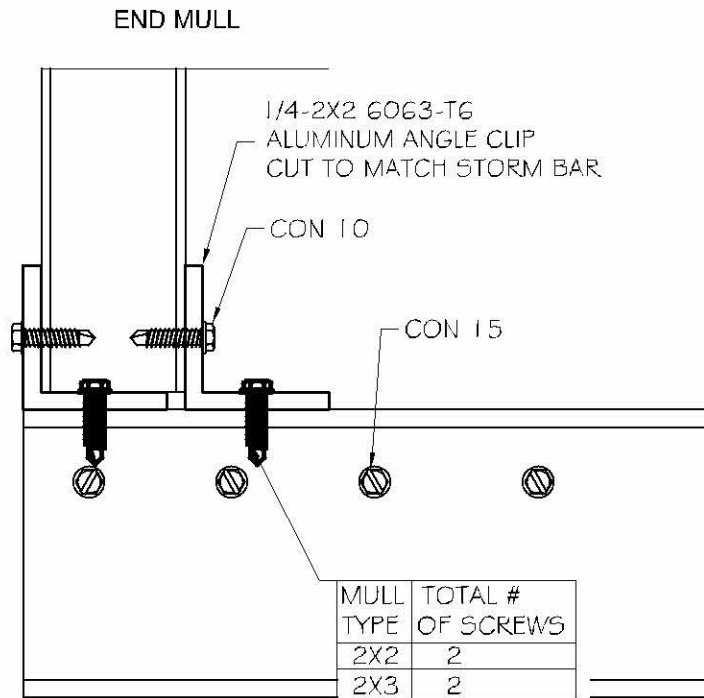
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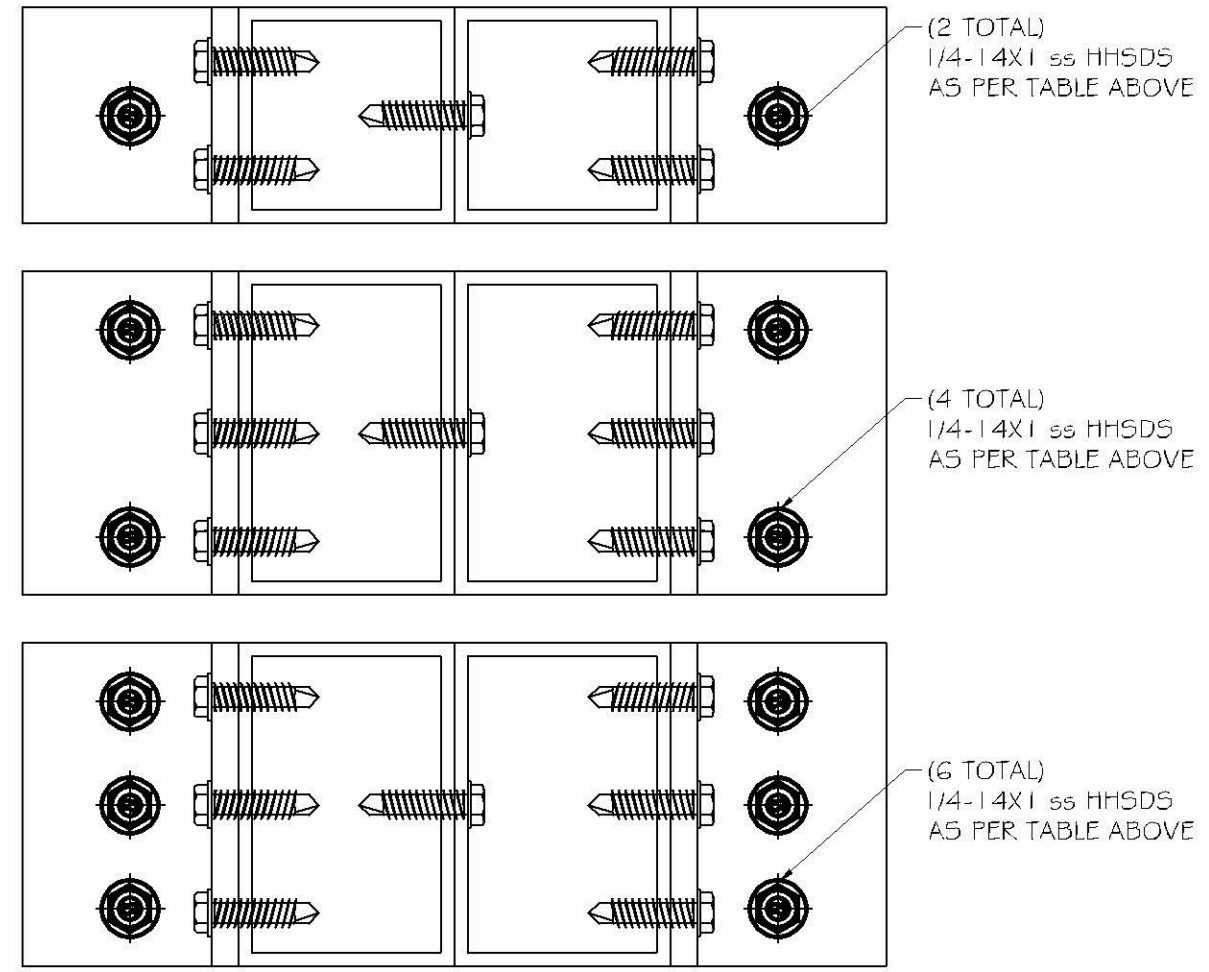
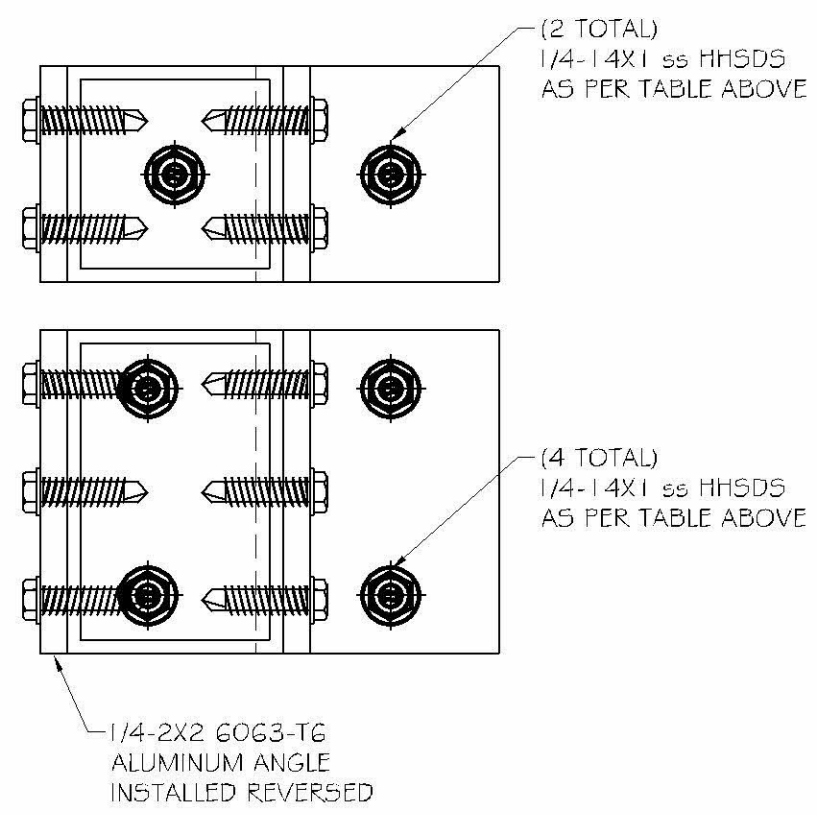
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CONNECTION 14
CLIP ANCHORED TO BUILT-OUT TUBE



ANCHORS TO BE DESIGNED BASED ON THE MINIMUM MULLION SIZE SELECTED FROM TABLES 7 AND 8.



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LICENSE
No. 52609
STATE OF
FLORIDA
PROFESSIONAL ENGINEER
Pedro De Figueiredo
PE 52609
09/16/2023

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PRODUCT:
E555 - ES44
EXTRUDED ALUMINUM ROLLING SHUTTER
LARGE & SMALL MISSILE
IMPACT RATED

MANUFACTURER:
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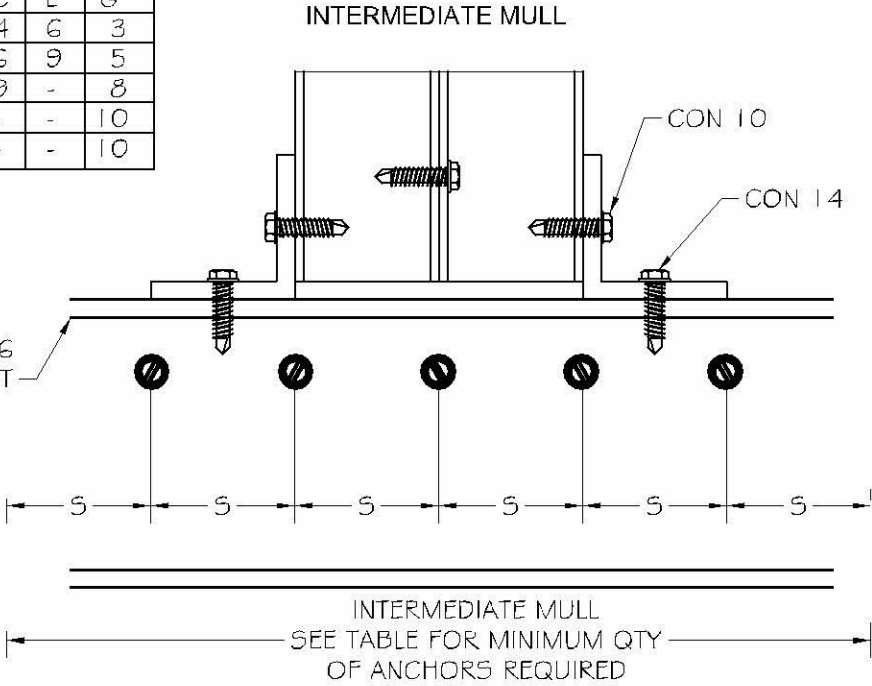
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CONNECTION 15
BUILT-OUT WALL ANCHORAGE
SUPPORTING MULLIONS

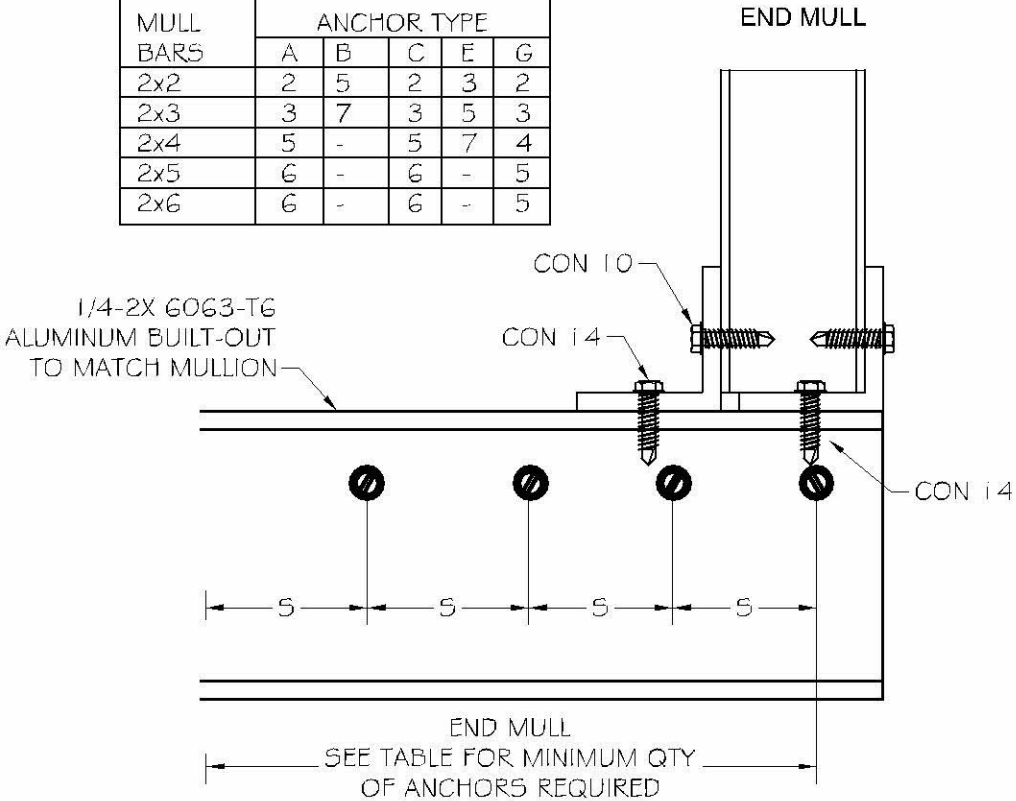
CONNECTION 15 -- INTERMEDIATE MULL
MIN. NUMBER OF ANCHORS
TO SECURE BUILT-OUT TUBE TO WALL

MULL BARS	ANCHOR TYPE				
	A	B	C	E	G
2x2	4	10	4	6	3
2x3	6	-	6	9	5
2x4	9	-	9	-	8
2x5	-	-	-	-	10
2x6	-	-	-	-	10



CONNECTION 15 -- END MULL
MIN. NUMBER OF ANCHORS
TO SECURE BUILT-OUT TUBE TO WALL

MULL BARS	ANCHOR TYPE				
	A	B	C	E	G
2x2	2	5	2	3	2
2x3	3	7	3	5	3
2x4	5	-	5	7	4
2x5	6	-	6	-	5
2x6	6	-	6	-	5



ANCHORS TO BE DESIGNED BASED ON
THE MINIMUM MULLION SIZE SELECTED
FROM TABLES 7 AND 8.

	ANCHOR TYPE			
	A	C	E	G
S	3.0	3.0	4.0	4.0
ED	2.5	1.0	4.0	4.0

S - ANCHOR SPACING
ED - EDGE DISTNCE

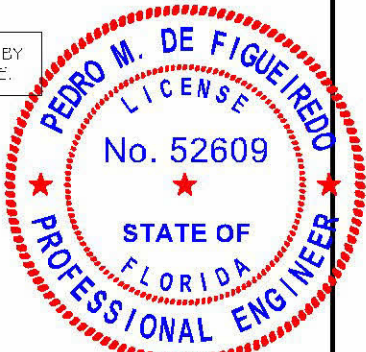
ANCHOR SCHEDULE

TYPE	DESCRIPTION	MANUFACTURER	EMBEDMENT	ED	BASE MATERIAL
A	1/4" ss TAPCON	ITW	1 3/4"	2 1/2"	3000 psi CONCRETE
C	1/4"-14 ss HHSMs	GENERIC	1 1/2"	1"	WOOD (SG >=.4G)
E	5/16" TAPCON	ITW	1 1/4"	4"	HOLLOW BLOCK
G	5/16" TAPCON	ITW	2 1/4"	2 1/2"	3000 psi CONCRETE

ANCHOR NOTES:

- 1- 1/4" TAPCONS CAN BE USED AS TYPE C INTO WOOD
- 2- 1/4" CARBON STEEL TAPCONS CAN BE USED AS AN ALTERNATE TO THE STAINLESS STEEL
- 3- METAL STRUCTURES ARE MADE OF MINIMUM 6063-T6 ALUMINUM OR ASTM A36 STEEL

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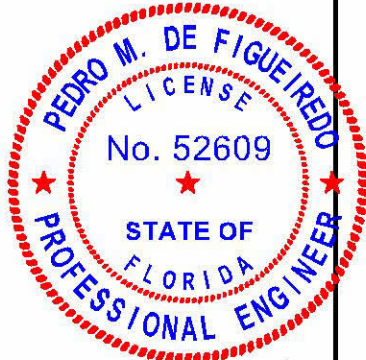
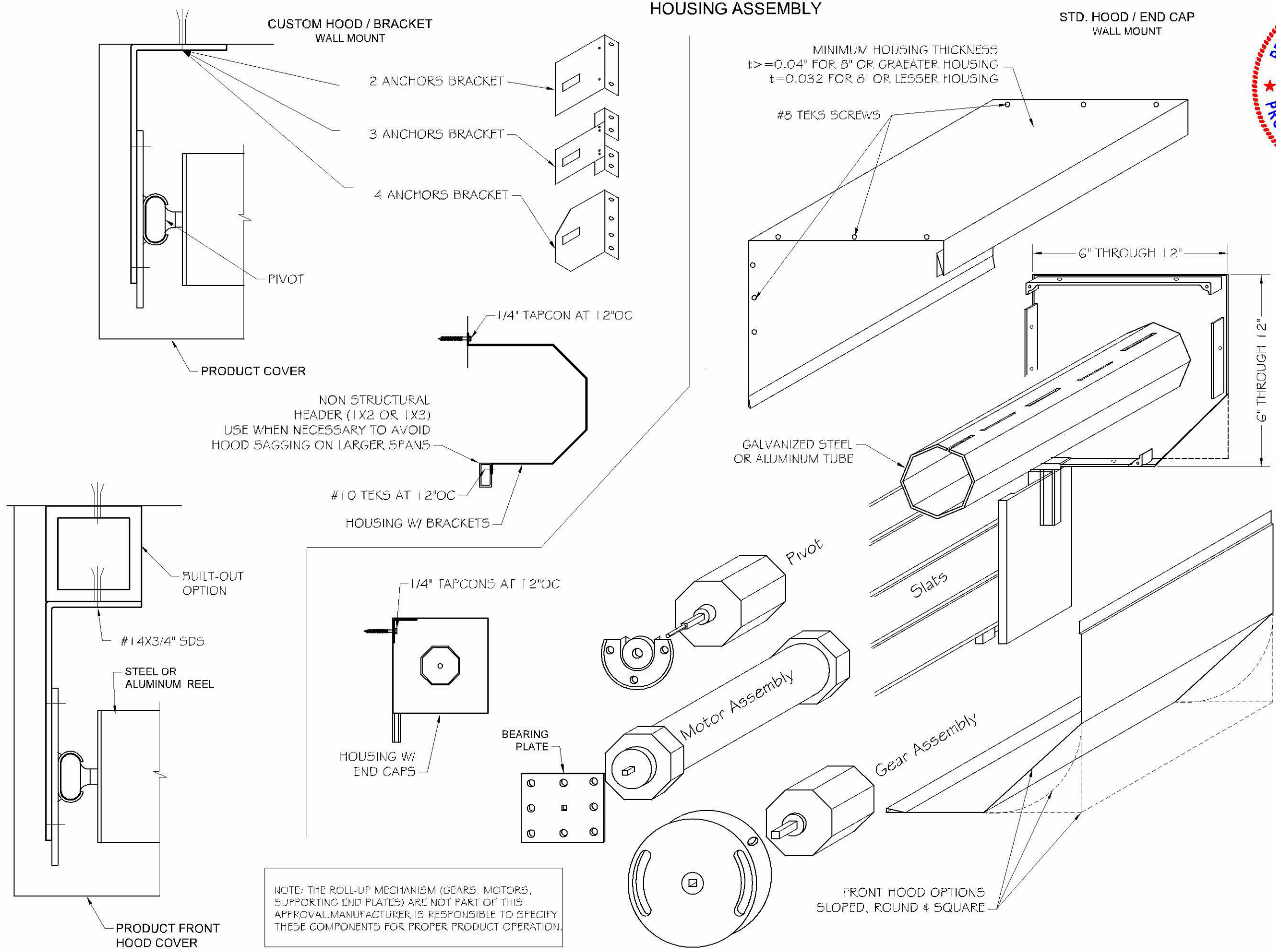


PRODUCT:
ES55 - ES44
EXTRUDED ALUMINUM ROLLING SHUTTER
LARGE & SMALL MISSILE
IMPACT RATED

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HOUSING ASSEMBLY



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LARGE & SMALL MISSILE
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